Top Researchers on Scientific Committees: Decision Quality and Opportunity Costs

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Abstract

Science disproportionately relies on the most productive researchers to evaluate the work of others, potentially diverting their scarce time from other tasks. We study this trade-off in Italy's national academic qualification system, where evaluators are randomly assigned to committees. Committees including better-published researchers make more informative selection decisions, particularly for more junior applicants: their qualified candidates have stronger subsequent citation and career outcomes, while their rejected candidates perform worse. However, these benefits come with substantial opportunity costs. Committee service reduces evaluators' publication output and PhD supervision, with larger research losses among more productive researchers and those working in smaller teams.

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1 Introduction

Peer evaluation of research is central to scientific progress. By allocating journal space, funding, promotions, and recognition, it affects the direction of science and provides incentives to produce impactful, high-quality research. However, expert evaluation takes substantial time from researchers. The journal peer review system alone is estimated to consume more than 100 million hours annually.[1]. This evaluation burden tends to fall disproportionately on a small group of highly qualified and productive scientists whose time may be particularly valuable elsewhere.[2] At the same time, demand for evaluation appears to be increasing, with journals, funding agencies, and other scientific organizations reporting growing difficulty recruiting evaluators, raising concerns about the sustainability of a system that relies heavily on scarce expert time.[3]

Despite these concerns, there is surprisingly little empirical evidence on the extend to which more productive researchers improve the quality of scientific evaluation and whether they incur greater opportunity costs when they serve. Neither relationship is theoretically clear. The ability to recognize novel and impactful work may not necessarily require the ability to produce it. Yet researchers working closer to the frontier may be better able to distinguish high-potential research from work that is merely competent. The costs are similarly ambiguous. An hour devoted to evaluation may displace more research and mentoring when spent by a more productive researcher. However, more productive researchers may also need less time to reach their decisions, which could offset the higher cost of each hour.

In this paper, we provide causal evidence on the benefits and the costs of assigning better-published researchers to evaluation committees. We ask whether evaluators with stronger publication records improve committee decisions; whether these gains arise solely from their own contributions or also through their influence on fellow members' judgment and effort; and whether committee service leads to larger reductions in publication output, PhD supervision, and willingness to serve

1. Aczel et al. (2021) estimate that researchers spend more than 100 million hours on peer review, worth over $2.5 billion when valued at average hourly wages in 2020.

2. Survey evidence shows that peer review is highly concentrated in the hands of a few scientists (Kovanis et al., 2016; Publons, 2018; Vesper, 2018). Kovanis et al. (2016) estimate that 20% of researchers perform between 69% and 94% of biomedical peer reviews. Similarly, Publons (2018) estimate that 10% of reviewers perform 50% of reviews.

3. *Nature*'s coverage of the Publons Global State of Peer Review, based on a survey of more than 11,000 researchers, reports "reviewer fatigue," difficulty finding reviewers, and rising invitations per completed review (Publons, 2018; Vesper, 2018). A UKRI-commissioned review similarly highlights reviewer burden, weak incentives, and the need to sustain reviewer supply for grant evaluation (Kolarz et al., 2023). The NIH recently simplified the review framework for most research-project grants, in part to reduce the growing burden on reviewers and refocus their attention on scientific and technical merit (NIH, 2024). Surveys also identify time burden as a major obstacle to recruiting and retaining editorial board members (Funk et al., 2024).

again. We study these questions using data from Italy's national academic qualification system, the *Abilitazione Scientifica Nazionale* (ASN). In this system, national discipline-level committees act as a first-stage filter for promotions: candidates seeking associate or full professor positions must obtain a national qualification before they can be appointed by a university. Committee members are drawn at random from the pool of eligible professors who volunteer to serve, subject to known institutional constraints. This random assignment allows us to address all three questions causally. First, we study selection quality by comparing outcomes of candidates evaluated by committees whose members have, by luck of the draw, stronger or weaker publication records. Second, we study peer dynamics by exploiting that, conditional on being drawn, the quality of one's committee peers is itself randomly assigned. Finally, we estimate the opportunity costs of service by comparing professors drawn to serve with professors who had the probability to be selected but were not drawn.

We use information on around 10,000 professors in the pool of potential evaluators, of whom about 3,200 served on more than 680 committees across 184 disciplinary areas between 2012 and 2021. To study committee decisions and peer effects, we focus on the approximately 70,000 applications from the first wave of the 2012 evaluation cycle, when candidates applied before committees were formed. For each candidate, we observe the individual report and vote of each of the five committee members. We link these records to an administrative database covering all professors at Italian public universities, as well as to bibliographic data and PhD dissertation records. Together, these sources allow us to track candidates' research output and career progression, as well as evaluators' research output and PhD supervision.

We first show that better-published committees are more selective and place more weight on candidates' publication impact and less on their publication quantity. Replacing an average evaluator with one whose publication record is one standard deviation stronger lowers the qualification rate by about 1.4 percentage points, against a baseline of 37%.

Better-published committees not only select different candidates but they also make more informative selection decisions: the candidates they qualify subsequently receive more citations, while those they reject receive fewer. Replacing an average evaluator with one who is one standard deviation better-published increases the citation gap between qualified candidates and the average applicant in the same examination by roughly 7%. It also widens the gap between the examination mean and rejected candidates' average future citations by roughly 6%. Greater strictness alone, with no change in candidate rankings, would predict a different pattern: marginal candidates

would move from the qualified to the rejected group, raising average outcomes in both groups. This pattern holds for subsequent citations to work published both before and after the evaluation, suggesting that more productive committees are better at recognizing not only the value of candidates' existing contributions but also the potential of their research agendas.

Research expertise matters most when committees assess junior researchers. Among associate-professor applicants, replacing an average evaluator with one whose publication record is one standard deviation stronger raises the citation advantage of qualified candidates by about 10% and widens the shortfall of rejected candidates by about 7%. The corresponding estimates for full-professor applicants are close to zero and statistically insignificant. This contrast suggests that research expertise is most valuable when candidates' shorter track records provide noisier signals of their potential. For associate-professor applicants, subsequent promotion to full professor provides another measure of selection quality, based on evaluations of an expanded research record by different committees later in their careers. The same replacement raises qualified candidates' probability of reaching this rank within ten years by roughly one percentage point, or 6% relative to the 17% baseline.

Better-published researchers affect committee decision-making not only through their own votes but also through influence on their peers. Evaluators randomly assigned to serve with better-published peers vote more like them: they become more selective and place greater weight on candidates' publication impact. They also write longer reports using less common vocabulary, suggesting that they exert more effort. This is consistent with reputation concerns within committees (Swank and Visser, 2023), while the convergence in voting may also reflect persuasion or deference to more accomplished colleagues.

The benefits of involving accomplished researchers in evaluation, however, come at a cost to their own research and doctoral supervision. Committee service reduces publication output within the following four years by approximately 0.05 standard deviations, equivalent to 28% of a typical year's output. When we distinguish publications by journal impact, losses in lower-impact journals remain statistically significant through the end of the observation period, with no evidence of catch-up. For high-impact publications, estimated losses are smaller when output is measured over longer periods, consistent with delays. Losses are larger in the Social Sciences and Humanities and smaller among evaluators who work in larger teams, which may partly reflect coauthors taking on additional work and thus bearing some of the costs of committee service. Committee service also persistently reduces PhD supervision. The decline in completed dissertations becomes apparent about four years

after the draw, consistent with evaluators taking on fewer new students while serving. Committee service also discourages future participation: researchers assigned to a committee are less likely to volunteer to serve again when they next become eligible.

Better-published researchers incur larger research losses. Within four years of assignment, committee service reduces the output of an evaluator whose publication record is one standard deviation stronger by about 70% more than that of an average evaluator. Since such evaluators are only about 50% more productive before assignment, their losses are more than proportional to their output. Service also lowers their probability of volunteering again by 2.6 percentage points, nearly twice the effect for an average evaluator. Relying on better-published evaluators may therefore reduce not only current research output but also the future supply of expert evaluators. Aggregate participation over the following decade fits this pattern: top researchers are initially more likely to volunteer, but their participation declines and converges with that of other researchers.

Our findings relate to several strands of the literature. First, we contribute to research on the role of star researchers in shaping science. Prior work documents their influence through collaboration, training, and competition, as well as their effects on the entry of new researchers and ideas (Azoulay et al., 2010; Waldinger, 2010; Borjas and Doran, 2012; Waldinger, 2012; Azoulay et al., 2019). We study their role as scientific gatekeepers (Merton, 1957; Stephan, 2010), providing causal evidence that evaluators with stronger research records improve selection quality, as measured by candidates' subsequent research and career outcomes. The gains are concentrated at the associate professor level, consistent with expert judgment mattering more when publication records reveal less about future potential. Our focus on research achievement as a vertical dimension of expertise complements studies of its horizontal dimension, typically measured by research proximity between evaluators and the work they assess (Boudreau et al., 2016; Li, 2017; Bello et al., 2025; Krieger et al., 2025).[4]

Second, we provide causal evidence on the opportunity costs of scientific evaluation. Consistent with models of multitasking (Holmstrom and Milgrom, 1991) and evidence on professors' time allocation (Myers and Tham, 2026; Myers et al., 2026), we show that committee service displaces publication output and PhD supervision. Heterogeneity by productivity and team size indicates that these costs depend on whose time is diverted and how research is organized.

These costs inform debates over how much institutions should rely on expert judgment. In

4. Our finding that better-published evaluators place greater weight on research impact relative to publication counts is in line with evidence that candidates with publications in "predatory" journals are less likely to qualify when assessed by better-published committees (Bagues et al., 2019b).

academia, bibliometric indicators can reduce information frictions (Hager et al., 2024), but how far they can substitute for expert evaluation remains contested (Bertocchi et al., 2015; Stephan et al., 2017; Heckman and Moktan, 2020). Our findings highlight the trade-off involved: research accomplishment matters for selection even when evaluators observe detailed publication records, yet the researchers whose judgment improves decisions the most bear larger research losses from service. It is therefore important to identify where expert time adds the most value. These findings also inform broader debates over when tests and algorithms can complement or replace human judgment, including in hiring and child protection (Hoffman et al., 2018; Li et al., 2026; Baron et al., 2026). We complement this literature's focus on decision quality by measuring the opportunity costs of drawing on experts and showing that they are greater for those whose judgment is most valuable.

Third, we contribute to research on expertise in collective decisions by showing how accomplished members influence their peers. This complements evidence on within-committee responses to peer assessments (Lane et al., 2022) and the influence of seniority within teams (Chan, 2021). We provide causal evidence that better-published evaluators affect colleagues' votes and effort in report writing, even though all members hold the same academic rank and have equal voting rights. The positive effort response is informative about the relative strength of the competing forces discussed in theoretical work. Members may exert less effort when they can rely on better-informed colleagues (Persico, 2004) or when accomplished members are expected to receive most of the credit for decisions (see Visser and Swank, 2007). Conversely, the desire to demonstrate competence to accomplished peers may encourage effort (Swank and Visser, 2023). Our evidence is consistent with the latter force dominating.

The remainder of the paper is organized as follows. Section 2 describes the institutional setting of Italian academic promotions. Section 3 outlines our data sources. Section 4 presents results on selection outcomes and discusses the mechanisms through which better-published researchers affect decision-making. Section 5 presents results on the opportunity costs of committee service. Section 6 puts our findings in perspective through a back-of-the-envelope comparison of the benefits and costs of involving top researchers in evaluation and an approximation of the elasticity of research output with respect to time. Section 7 concludes with implications for policy.

2 Italian National Academic Qualifications

Italy introduced the National Scientific Qualification system, *Abilitazione Scientifica Nazionale* (ASN), in 2012 under the Gelmini Reform to harmonize academic standards across universities and

limit local favoritism. The system added a national qualification stage to university appointment procedures: researchers seeking associate or full professorships must first qualify through a field-level committee before universities can appoint them.

Committee members are drawn randomly from a pool of eligible volunteers. In the first cycle, candidates applied before learning who would evaluate them, and committees had substantial discretion over evaluation criteria. This setting allows us to examine how committee composition affects the evaluation of a predetermined applicant pool. We describe below how evaluators are selected and how committees assess candidates.

2.1 Evaluator Selection

Evaluation committees are formed at the field level and consist of five members.[5] Committees are allocated in three stages. First, the Ministry sets field-specific eligibility requirements based on research productivity. Second, eligible professors then decide whether to volunteer. Finally, committee members are selected from this pool through a public random draw, subject to institutional constraints.

Eligibility. Eligibility criteria reflect disciplinary publication norms. Full professors must exceed the median among all full professors in their field on at least two of three research productivity indicators defined by the Ministry. In STEM+M fields – which, under these procedures, include medicine and psychology along with Sciences, Technology, Engineering, and Mathematics – the indicators are journal publication counts, citation counts, and the H-index. In Social Sciences and Humanities (SSH), they cover journal publications, articles in high-impact journals, and books and book chapters. The specific thresholds and requirements vary slightly across evaluation cycles.

Volunteering. Professors volunteer by responding to a public call from the Ministry and submitting their curriculum vitae. The Ministry verifies their eligibility before admitting them to the pool of potential evaluators.

Draw procedure. The Ministry conducts a public computerized draw to select committee members.[6] Within each field, eligible volunteers are listed alphabetically and assigned an identifier. A

5. Official documents regulating the process are available at https://abilitazione.mur.gov.it/public/index.php?lang=eng (retrieved September 2025).

6. Procedures available at https://abilitazione.mur.gov.it/public/documenti/Modalita_sorteggio.pdf & https://abilitazione.mur.gov.it/public/documenti/2016/Allegato_1_Procedura_sorteggio_commissioni_

random sequence of these identifiers, drawn without replacement, determines the order in which volunteers are considered for appointment. Committee seats are filled subject to two constraints: no two members may be affiliated with the same university, and seats are reserved for large subfields before the remaining positions are filled. Volunteers whose appointment would violate the university constraint are skipped. If a selected evaluator resigns, a replacement is drawn under the same rules.

These constraints and the composition of the volunteer pool imply that researchers may have different probabilities of being assigned to a committee, which we account for in the empirical analysis below.

In the 2012 ASN cycle, committees generally comprised four evaluators affiliated with Italian universities and one based in another OECD country (drawn through an analogous randomization procedure), although a small number consisted entirely of Italy-based evaluators. In later cycles, the requirement for an international member was removed, and all five members were drawn from Italian universities. Our analysis focuses on evaluators affiliated with Italian universities to ensure comparability across cycles and because of concerns about limited engagement in the evaluation process among foreign committee members.[7]

Mandate and service conditions. Committee members serve for two years and evaluate all applications submitted during their term. After completing a term, they are ineligible to serve again for at least three years, a restriction we account for when examining future volunteering in Section 5. Italy-based evaluators receive no additional remuneration for committee service, although they may request partial reductions in their teaching duties.

2.2 Evaluation Process

Before assessing candidates' applications, each committee holds an initial meeting to establish its organizational rules and evaluation criteria. In the 2012 cycle, committees had substantial discretion in setting these criteria, including the weight assigned to different aspects of candidates' research records.[8]

`ASN_2016.pdf` (accessed June 27, 2025).

7. Contemporary commentary highlighted language barriers and practical difficulties associated with foreign members' participation (ROARS, 2014). In our data, foreign evaluators wrote reports that were, on average, 20% shorter, suggestive of lower engagement. Results are robust to including these evaluators, although estimates are less precise.

8. The Ministry recommended considering three bibliometric indicators similar to those used to determine evaluator eligibility. In later cycles, minimum bibliometric requirements were introduced, while committees retained discretion over additional criteria and candidates' overall assessment.

Evaluations are based exclusively on the material submitted by candidates: their curricula vitae and up to ten selected publications. Each committee member assesses each application within their field and prepares an individual written judgment. These assessments are discussed and finalized in subsequent committee meetings, during which the committee also records its final decisions. A positive qualification decision requires a favorable vote from at least four of the five members.

Committee service entails a substantial workload. Between 2012 and 2018, each committee evaluated approximately 270 candidates on average (Italian National University Council, 2023). Evaluators must establish evaluation criteria, write an individual report for each candidate, and participate in committee deliberations. Informal discussions with evaluators suggest that these duties may require up to 200–300 hours per member over a two-year term, although the burden likely depends on the number of applications and committees' evaluation and reporting practices.

The process is highly transparent: the Ministry publishes evaluators' and candidates' curricula vitae, individual votes, and evaluation reports online. The national evaluation agency also collects and publishes indicators of candidates' research output, comparing their records with discipline-specific benchmarks.

3 Data

This section describes our data sources, defines the main variables, and presents key descriptive statistics.

3.1 Data sources

Administrative data on Italian professors. We obtain administrative records on all professors employed at Italian public universities between 2010 and 2023 from the Ministry of Universities and Research.[9] The resulting panel covers approximately 90,000 distinct professors across 95 universities and records their university affiliation, department, research area, and academic rank.

National qualification system (ASN). We collect administrative records on the identity of evaluators across four ASN cycles between 2012 and 2021.[10] These records identify the pool of potential evaluators, the researchers initially selected by the random draw, and those who ultimately served following resignations and replacements.

9. The data are available at https://cercauniversita.mur.gov.it/, last accessed 6 October 2026.

10. The Ministry maintains a dedicated ASN website at https://abilitazione.mur.gov.it/public/index.php, last accessed 5 October 2026.

For the first wave of the 2012 cycle, we also use the database assembled contemporaneously from the Ministry's website by Bagues et al. (2017).[11] This database contains applicants' names, curricula vitae, and evaluators' individual reports for each candidate. The individual reports record each evaluator's vote on each candidate and allow us to construct text-based proxies for evaluative effort, including report length and linguistic complexity.

Bibliographic databases. We combine three sources to measure candidates' and evaluators' research before and after qualification. First, we extract candidates' publication records from the curricula vitae submitted with their ASN applications, which capture the information available to evaluators. Second, we collect publication records from universities' institutional repositories using IRIS, a research information system widely used in Italy. These records cover 2000–2024 and include journal articles, books, chapters, and other outputs. Third, we supplement IRIS with publication and citation data from OpenAlex (Priem et al., 2022), expanding coverage and reducing reliance on self-reported institutional records.

We link researchers to these databases using names, affiliations, and research fields, and remove duplicate publications across sources. For publications appearing in both databases, we use citation counts from OpenAlex. Together, IRIS and OpenAlex cover 96% of professors in our administrative data and approximately 92% of ASN candidates. Appendix B describes the matching and de-duplication procedures.

Doctoral graduates database. We collect records of PhD theses catalogued in the National Central Library of Florence (BNCF). Although doctoral theses in Italy are legally required to be deposited, their coverage in online catalogs depends on universities' adoption of electronic deposit procedures.[12] The BNCF records therefore capture digitally deposited and catalogued theses rather than all Italian doctoral theses.

We focus on 2012–2023, when coverage is substantially more complete, and retain institutions with at least five recorded theses in 2012 per 100 tenured faculty members. This decision is empirically motivated: institutions above this baseline threshold exhibit substantially more regular deposits in subsequent years, whereas those below it frequently have sporadic or near-zero records.[13]

11. Materials on applicants are available online for only six months after the decision, as opposed to information about evaluators.

12. See the BNCF documentation at
https://bncf.cultura.gov.it/servizi/deposito-delle-tesi-di-dottorato/.

13. Among institutions above the cutoff, the median annual number of recorded theses over 2012–2023 is 17.6 per

This restriction excludes approximately 43% of institutions in our sample.

3.2 Descriptive statistics

Evaluators. We estimate that approximately half of full professors met the eligibility thresholds across broad disciplinary groups, with a notably higher share of 83% in SSH in the 2012 cycle (Appendix Figure A1).[14] Among professors meeting these criteria, approximately 50% volunteered in 2012. This share declined to roughly 35% by 2021. The evaluator pool was positively selected through both eligibility requirements and voluntary participation (Appendix Table A1).

Across the four examination cycles, our data cover 685 evaluation committees in 184 fields and approximately 10,000 distinct full professors who met the eligibility requirements and volunteered for service. On average, the pool of potential evaluators consisted of 26 professors.

Approximately 9% of initially assigned evaluators resigned and were replaced through the random replacement procedure. Resignations are weakly related to observable publication records (Appendix Table A1, Panel C).

We characterize evaluators' research records using publications from the ten years preceding each exam. The average researcher in the pool of potential evaluators produced 60 publications over this period, approximately 90% of which were journal articles; the remainder comprised books, book chapters, conference proceedings, and other outputs (Appendix Table A2, Panel A). We classify journals as high-impact using field-specific criteria: articles in the top quartile (Q1) by Article Influence Score (AIS) in STEM+M fields and the Italian evaluation agency's A-journal classification in economics, business, and other SSH fields.[15] This approach accommodates differences in the availability of journal impact measures across fields, particularly their more limited coverage in SSH. Approximately 40% of journal articles by potential evaluators appeared in high-impact journals.

Our baseline measure of *evaluator quality* is the number of high-impact journal articles published in the ten years preceding the exam. It captures a readily observable dimension of research accomplishment that institutions can use when selecting evaluators.[16] Whether it predicts better evaluation is an empirical question central to our analysis. We standardize this variable to have

100 tenured faculty members, and the 25th percentile is 9.8. Among institutions below the cutoff, the corresponding figures are 1.3 and 0.5.

14. We do not directly observe the Ministry's eligibility determinations for all professors and therefore reconstruct eligibility using the official criteria.

15. AIS measures citation impact, weighting citations by the influence of the citing journal and excluding journal self-citations.

16. We use "quality" as shorthand for this publication-based measure; whether it predicts better evaluation is an empirical question.

mean zero and standard deviation one within discipline-year cells among all full professors at Italian public universities. We also create indicators for *top researchers*, defined as those in the top 25%, top 10%, 5%, and 1% of this distribution.

Our baseline measure of *committee quality* is the average standardized quality of its Italy-based members. We later examine whether outcomes depend on the number of top researchers and the threshold used to define them. Appendix C assesses robustness to alternative measures of research accomplishment.[17]

Among potential evaluators, professors meeting the top-quartile definition produced an average of 86 publications in the preceding ten years, compared with 44 publications among other professors (Appendix Table A2, Panel A). The two groups worked in teams of similar size, averaging about six authors per paper.[18]

In order to measure the opportunity costs of committee service, we track evaluators' post-assignment research productivity and student supervision. In the four-year window starting with the assignment year, the average researcher in the pool of potential evaluators produced 28 publications, roughly one third of which were high-impact articles, and supervised 2 PhD graduates (Appendix Table A2, Panel B). Over the nine-year period beginning in the assignment year, these totals were approximately twice as large (Panel C).

Candidates. The first wave of the 2012 ASN cycle received 69,020 pre-registered applications from more than 46,000 researchers, with an average of 375 applications per field. Participation was widespread: approximately 61% of Assistant Professors and 60% of associate professors employed at Italian universities at the end of 2012 applied.[19] Roughly one third submitted multiple applications, either across related fields or for both ranks within the field.

Candidates could withdraw after committees were drawn and evaluation criteria were announced. The database assembled by Bagues et al. (2017) contains the curricula vitae of all pre-registered candidates, published before the final committee membership was confirmed. Approximately 14% of pre-registered applications were withdrawn, leaving 59,151 applications for evaluation (Italian National University Council, 2023). There is no evidence that withdrawal

17. The qualitative patterns are consistent across measures, although statistical significance varies.

18. We winsorize the number of authors at the 99th percentile to limit the influence of exceptionally large teams. The median number of authors per paper, which is unaffected by this adjustment, is 4.4 for top researchers and 4.2 for other researchers.

19. These participation rates are calculated using Ministry records on Assistant Professors (*Ricercatori*) and associate professors (*Associati*) employed as of December 31, 2012.

was systematically related to committee research quality (Appendix Table A3). Qualification was awarded to 37% of pre-registered applications, equivalent to 43% of evaluated applications. Two thirds of applicants were already employed at an Italian university.

We use candidates' curricula vitae to measure pre-evaluation research profiles. We summarize candidates' research profiles at the time of the evaluation along two dimensions: *quantity* and *impact*. These measures are not intended to provide an exhaustive description of candidates' research profiles, but to capture two salient and widely discussed dimensions of research performance along which we can assess systematic differences in evaluation decisions. We measure *quantity* as the total number of academic outputs. Our measure of *impact* is the share of journal articles published in high-impact journals, using the same field-specific classifications as above: Q1 journals by AIS in STEM+M and A-journals in SSH. Expressing *impact* as a share rather than a count reduces the concern that it captures nonlinear effects of publication volume. Empirically, these measures predict qualification outcomes more strongly than a range of alternative candidate characteristics (Appendix C.3).

Publication patterns vary substantially across fields (Appendix Table A4). In STEM+M, the mean share of journal articles published in Q1 journals ranges from 30% to 50% across fields. In economics and business, the corresponding share is approximately 15%, while 21% of articles appear in A-journals. In the remaining SSH fields, the mean share of A-journal articles is approximately 29%. To account for differences in publication practices and journal classifications, we standardize both quantity and impact to have mean zero and standard deviation one within the relevant candidate pools: within field and application level.

Approximately 15% of candidates had a coauthor or a colleague from their institution on their committee.[20] For candidates employed at Italian universities, we observe their subfield (*settore scientifico-disciplinare*) of appointment. We also measure research similarity using the topics of candidates' and evaluators' publications in OpenAlex. The topics are sparse, since there are approximately 4,500 OpenAlex topics, therefore we reduce the dimensionality of the topic vectors to 50 components and compute cosine similarity between each candidate–evaluator pair in this reduced space.

Finally, we track candidates' subsequent research and career outcomes using the combined OpenAlex and IRIS publication data and CINECA appointment records. Qualified candidates

20. As Bagues et al. (2019a) explain, coauthors and institutional colleagues were not formally subject to conflict-of-interest rules in the first ASN round. Committees could impose their own restrictions, but only three required abstention, affecting 84 candidates who consequently received four reports rather than five.

produced 62 publications over the ten years following the evaluation, their pre-evaluation publications received approximately 1,000 subsequent citations, and their post-evaluation publications received approximately 1,500 citations (Appendix Table A5). Rejected candidates were about half as productive.

Individual evaluation reports. Individual evaluation reports provide the evaluator's qualitative assessment of the candidate and conclude with a recommendation on qualification. Our dataset contains 241,744 reports written by Italian evaluators.[21] Among reports by Italy-based evaluators, 45% recommend qualification (Appendix Table A6). Agreement within committees is high: approximately 86% of qualification decisions are unanimous.

We proxy evaluator effort using report length (word count) and lexical rarity, measured by inverse document frequency (IDF), which gives greater weight to words appearing in fewer reports. Before calculating IDF, we remove punctuation, numerals, stop words, and first names, and stem words to reduce inflectional variation, including Italian gendered forms. *Average IDF* measures mean lexical rarity, while *Total IDF* sums word weights, combining rarity and text volume. Reports average 182 words and an Average IDF of 2.54 (Appendix Table A6). STEM+M reports are shorter than SSH reports (150 versus 233 words), use more common vocabulary (Average IDF of 2.29 versus 2.94; Total IDF of 220 versus 438), and mention bibliometric indicators slightly more often (47% versus 43%). We standardize all textual features to have mean zero and standard deviation one within 86 macro-fields and application level.[22]

We perform two exercises to assess whether these textual features are likely to be good proxies of evaluator effort by examining connections and borderline decisions. Reports are longer and use less common vocabulary when evaluators are connected to candidates, including through prior coauthorship, conditional on candidate and evaluator fixed effects (Appendix Table A7). The same pattern holds for candidates whose qualification hinges on a single vote, conditional on candidate publication quantity and impact and evaluator fixed effects (Appendix Table A8).

21. Some committees included five Italy-based evaluators, so the number of domestic reports can exceed four per evaluated application. Reports are missing for 202 applications because of a technical issue.

22. Normalizing at a broader level preserves between-examination variation, which is needed to distinguish peer effects from own-productivity effects: within an examination, higher-productivity evaluators mechanically have lower-productivity peers.

4 Decision Quality and Peer Effects

In this section we examine how committee composition affects selection quality, how evaluators influence their peers, and whether returns to expertise are nonlinear.

4.1 Committee composition and selection outcomes

We assess decision quality using candidates' subsequent research performance and career progression. Committees observe past research output but must also infer less readily measured dimensions of potential, such as the originality of candidates' research agendas and the likely impact of their work. If better-published committees interpret these signals more accurately, the candidates they qualify should perform better in the future.

Stronger outcomes among qualified candidates, however, need not establish better screening. Better-published committees may simply impose a higher qualification threshold, selecting a stronger group by qualifying fewer applicants.

We therefore proceed in two steps. First, we examine whether better-published committees are more selective and weigh observable candidate characteristics differently. Second, we compare subsequent outcomes among both qualified and rejected candidates. As we explain below, considering the two groups together helps distinguish more effective screening from greater stringency.

Committee quality and the qualification decision. We first examine whether better-published committees are more selective and whether their qualification decisions place different weight on candidates' publication quantity and impact. We estimate:

$$Q_{i,e} = \beta_0 + \beta_1 T_e + \beta_2 N_i + \beta_3 I_i + \beta_4 (T_e \times N_i) + \beta_5 (T_e \times I_i) + \mathbf{X}'_{i,e}\mathbf{\Gamma} + \varepsilon_{i,e} \tag{1}$$

where $Q_{i,e}$ is an indicator for whether candidate i qualified in examination e, T_e, is committee quality, N_i is candidate publication quantity, and I_i is candidate publication impact.

Because candidate measures are centered within examinations, β_1 captures the effect of committee quality on qualification for a candidate with average publication quantity and impact. Evaluator quality is standardized among full professors in the same field, so a one-unit increase in committee quality corresponds to raising every member's quality by one standard deviation. The interaction coefficients, β_4 and β_5, capture how the relationship between qualification and each publication measure changes with committee quality.

Our identification strategy relies on variation in committee quality induced by the random assignment of evaluators. Given the features of assignment mechanism, some committees may be stronger than others in expectation, hence, we follow Borusyak and Hull (2023) to isolate the random component of the initially drawn committee's quality:

$$\widetilde{Z}_e = Z_e - \mathbb{E}[Z_e \mid \mathcal{W}_e]$$

where Z_e is the quality of the initially drawn committee and $\mathcal{W}_e$ comprises the eligible evaluator pool and the restrictions governing the draw. We calculate expected committee quality by simulating the assignment algorithm one million times. Thus, $\widetilde{Z}_e$ measures how much the initially drawn committee's quality differs from its expectation under the assignment mechanism. We instrument T_e, $T_e \times N_i$, and $T_e \times I_i$ with $\widetilde{Z}_e$, $\widetilde{Z}_e \times N_i$, and $\widetilde{Z}_e \times I_i$, respectively.

The vector of controls includes broad-discipline indicators (Sciences; Medicine and Veterinary; Engineering; Social Sciences and Humanities) and measures of candidate–committee connections and research proximity. These measures are the number of evaluators from the candidate's institution, the number of prior coauthors, the number in the same subfield, and average research topic similarity. Each of these variables enter as their value following the initial committee draw net its expected value under the assignment mechanism. Controlling for these randomly induced differences helps distinguish changes in how committees weigh publication quantity and impact from effects associated with professional ties (Zinovyeva and Bagues, 2015; Colussi, 2018; Bagues et al., 2019a) or research proximity (Li, 2017).

We estimate Equation (1) by two-stage least squares (2SLS) and cluster standard errors at the examination level. As expected under random assignment, predetermined candidate characteristics are balanced across the resulting variation in committee quality (Appendix Table A9), and, consistent with a low resignation rate, initial committee quality strongly predicts the quality of the evaluating committee (Appendix Table A10).

Table 1 reports the estimation results. Better-published committees are more selective: replacing an average evaluator with one whose publication record is one standard deviation stronger reduces the qualification rate by about 1.4 percentage points, or roughly 4 percent relative to the baseline rate of 37 percent (Column 1).[23] Committee quality also changes how qualification

23. Because committee quality averages the standardized output of four Italy-based members, replacing one average member with an evaluator whose pre-exam publication record is one standard deviation stronger raises T_e by 1/4. We report committee-quality effects on this scale.

outcomes relate to observable candidate characteristics. For committees with better-published evaluators, qualification is less strongly associated with publication quantity and more strongly associated with publication impact (Column 2).[24] Improving the quality of one member by one standard deviation reduces the association with publication quantity by about 5 percent and increases that with publication impact by about 9 percent. These patterns are similar in STEM+M and SSH fields (Columns 3 and 4).

This shift toward publication impact does not appear to reflect greater mechanical reliance on bibliometric measures. Reports written by better-published evaluators are, if anything, less likely to mention citations, the h-index, or publications in top journals (Appendix Table A11).

Despite fewer qualifications, we find no significant decline in promotion rates within five years for candidates with average publication quantity and impact (Columns 5–8). Promotions nevertheless become more strongly associated with prior publication impact, suggesting that national committee composition influences who advances even though universities make the final appointment decisions.

Committee quality and the quality of selection. We next examine whether better-published committees identify candidates with stronger future prospects. We estimate

$$y_{i,e,t+k} = \alpha^q + \beta^q T_e + \mathbf{X}'_{i,e}\mathbf{\Gamma}^q + u^q_{i,e,t+k} \quad \forall\, q \in \{0,1\} \tag{2}$$

separately for candidates who qualify ($q = 1$) and who are rejected ($q = 0$). The outcome variable $y_{i,e,t+k}$ measures the research output and career advancement of candidate i, who participated in examination e, k years from the decision in year t.

The vector of controls includes four broad-discipline indicators; candidates' publication quantity, publication impact, and citations prior to the evaluation; indicators for candidates' affiliation and academic rank; and measures of candidate–committee connections and research proximity.[25] Conditioning on candidates' observable records allows us to test whether better-published committees identify dimensions of quality beyond those captured by bibliometric indicators, such as the novelty and potential impact of research agendas. This helps assess what expert judgment adds

24. Appendix Table C7 assesses sensitivity to our choice of committee quality measure using alternative indicators, including article and citation counts. The same qualitative pattern holds across these measures: committees with stronger research records place less weight on candidate publication quantity and more weight on publication impact.

25. These specifications allow the coefficients on control variables to differ between qualified and rejected candidates. Results are robust to a pooled specification that includes qualification status and its interaction with committee quality, while imposing common coefficients on the control variables. We also estimate specifications excluding controls for candidate observables; the results are qualitatively similar but less precise.

beyond information that could be incorporated into formal evaluation rules.

We estimate this equation by 2SLS, instrumenting T_e with $\tilde{Z}_e$, and cluster standard errors at the examination level. The coefficients $\beta^{q=1}$ and $\beta^{q=0}$ capture how subsequent outcomes vary with committee quality among qualified and rejected candidates, respectively.

To distinguish better selection from greater stringency, we examine subsequent outcomes among both qualified and rejected candidates, following research on medical decision-making (see Chan et al., 2022). Holding the ranking of candidates fixed, a higher qualification threshold should improve average subsequent outcomes in both groups. Raising the threshold moves the weakest qualified candidates into the rejected group. Because these candidates are stronger than those already rejected, average quality rises in both groups. Better screening can instead identify stronger candidates to qualify without improving the average quality of those rejected. We therefore interpret the joint pattern

$$\beta^{q=1} > 0 \qquad \text{and} \qquad \beta^{q=0} \leq 0$$

as consistent with improved selection.

Interpreting this pattern as evidence of better selection requires two further conditions. First, the outcomes must not be driven by differences in the returns to qualification across candidates selected at different thresholds. In Appendix D, we perform a Marginal Treatment Effect analysis where we exploit random variation in candidate–committee connections and research proximity that shifts the probability of qualification. Conditional on candidates' pre-evaluation productivity, this variation allows us to estimate the effects of qualification on subsequent outcomes for candidates who require more or less favorable committee draws to qualify. The estimates suggest that differences in gains from qualification are unlikely to be an explanation for the patterns we document. Second, committee composition must affect candidates' subsequent outcomes only through the qualification decision. Direct effects through feedback, as in journal peer review, are unlikely in this setting: the average committee assessed 375 applications without interacting directly with candidates, and evaluation reports were short and generally justified decisions rather than offered individualized advice. Moreover, serving on a national qualification committee confers no role in subsequent university-level promotion decisions, which are made by separate committees.

Table 2 reports the estimation results from Equation (2). Candidates qualified by better-published committees receive significantly more citations in the following ten years, whereas those rejected receive fewer (Column 3). These differences emerge despite small and statistically insignif-

icant differences in subsequent total publication output and the number of articles published in high-impact journals (Columns 1–2). Better-published committees therefore appear to distinguish candidates with research that will have greater impact, even though their subsequent publication records are similar in quantity and journal placements. Quantitatively, replacing an average evaluator with one whose publication record is one standard deviation stronger raises average future citations among qualified candidates by approximately 0.011 standard deviations and lowers them among rejected candidates by approximately 0.005 standard deviations. Qualified candidates are, on average, 0.15 standard deviations above the examination mean, implying an increase to their citation advantage by approximately 7%. Similarly, the decline among rejected candidates widens their shortfall below the examination mean by roughly 6%.

Interestingly, the citation advantage is present for both pre- and post-exam publications, and if anything, it is larger for post-exam works (Columns 4–5). Better-published committees therefore appear to be more able not only in recognizing the value of existing research more accurately, but also in identifying candidates with stronger subsequent trajectories.

The improvement in selection is concentrated among candidates applying for associate professorship. Better-published committees qualify associate professor candidates with substantially stronger subsequent citation outcomes and reject candidates with weaker outcomes, whereas the corresponding estimates for full professor candidates are small and statistically not different from zero. This pattern is consistent with evaluator expertise being more valuable when assessing researchers at an earlier career stage, whose shorter research records may provide fewer observable signals of future performance.

Table 3 provides additional evidence on selection quality by examining future career outcomes of associate professor candidates. Replacing an average committee member with an evaluator whose publication record is one standard deviation stronger raises the share of qualified associate-professor applicants who obtain a promotion to full professor within ten years by about one percentage point, or 5.9% relative to the baseline (Column 1). A similar pattern emerges for entry into future ASN evaluator pools: candidates qualified by better-published committees are more likely to be eligible and volunteer as evaluators in later cycles (Columns 2–3). Replacing an average committee member with a researcher whose publication record is one standard deviation stronger raises qualified candidates' probability of entering the evaluator pool by 0.75 percentage points (7.5%) and of entering as a top researcher by 0.25 percentage points (6.3%). Committee composition thus shapes not only current selection but also future evaluator pools.

We complement this analysis with a different strategy for distinguishing selection quality from stringency: we compare candidates qualified by committees with positive evaluator-quality shocks with candidates unanimously qualified by committees with negative shocks. Restricting the latter group to unanimous approvals yields nearly identical selection rates: the two groups represent 34.1% and 33.8% of their respective applicant pools. Despite this similarity in selectivity, candidates in the first group subsequently receive more citations and, among associate professor applicants, are more likely to become full professors (Appendix Table A12).

Together, these patterns support more informative selection by better-published evaluators, with gains concentrated among associate-professor applicants, even within a pool of full professors meeting demanding research requirements.

4.2 Peer influence in committee decision-making

We next examine the mechanisms through which better-published evaluators shape committee decisions: they may influence outcomes directly through their own assessments or indirectly by changing their colleagues' evaluation criteria and effort. We distinguish these channels using data on votes and textual characteristics of evaluation reports from evaluator–candidate level evaluation reports.

Experts' influence on voting. To assess whether the better-published researchers apply different criteria and whether their presence affects other evaluators' behavior, we estimate the following equation:

$$y_{i,j,e} = \beta_0 + \beta_1 T_j + \beta_2 T_{e,-j} + \mathbf{X}'_{i,j}\mathbf{\Gamma} + \varepsilon_{i,j,e}. \tag{3}$$

Here, $y_{i,j,e}$ indicates whether evaluator j votes to qualify candidate i, T_j measures the evaluator's own quality, and $T_{e,-j}$ is the mean quality of the other Italy-based committee members. We instrument realized peer quality with

$$\widetilde{Z}_{e,-j} = Z_{e,-j} - \mathbb{E}[Z_{e,-j} \mid \mathcal{W}_e, \mathit{drawn}_{j,e} = 1],$$

where $Z_{e,-j}$ is peer quality under the initial draw. Expected peer quality is calculated from one million simulated draws, conditional on evaluator j being selected, the pool of potential evaluators and the draw rules. Thus, identification uses random variation in the quality of peers assigned to

a given evaluator.

The coefficient β_1 captures the association of evaluation outcome with own quality conditional on peer quality; $\beta_2/3$ measures the effect of replacing one of the three Italy-based peers with a researcher one standard deviation better-published.

The vector of controls includes candidate productivity, measures of evaluator–candidate connections and research proximity. We estimate Equation (3) by 2SLS, clustering standard errors at the examination level.

Table 4 reports the results. Better-published evaluators tend to be stricter. A one standard deviation increase in own quality is associated with a 2.3 percentage point decrease in the probability of casting a positive vote (Column 1). We find that peer quality also affects voting: replacing one average peer with an evaluator whose publication record is one standard deviation stronger reduces an evaluator's probability of voting positively by approximately 1.8 percentage points.

In Column 2, we include candidate fixed effects and find that, on average, evaluators of different productivity do not vote differently for the same candidate. Together with Column 1, this pattern suggests that better-published peers shift committee-wide standards and induce convergence in voting rather than simply casting more negative votes themselves.

Next, we examine how votes relate to candidate characteristics. Better-published evaluators' votes are less strongly associated with publication quantity and more strongly associated with publication impact (Column 3). Peer quality leads to a similar shift: replacing a peer with another whose publication record is one standard deviation stronger reduces the marginal association with publication quantity by about 0.6 percentage points and increases that with publication impact by about 1 percentage point (Column 4).

Experts' influence on effort. Better-published peers may influence not only evaluators' judgments but also the effort they devote to assessment. We examine this possibility by estimating Equation (3) using proxies for evaluator effort – evaluation report length, average IDF, and total IDF – as the outcome variables.

Table 5 shows no systematic relationship between an evaluator's own quality and the measures of effort, conditional on peer quality. By contrast, we find that an increase in the productivity of peers leads to an increase in all three proxies of effort. Replacing an average peer with another whose publication record is one standard deviation stronger raises effort by 0.06–0.07 standard deviations, depending on the measure.

Taken together, we find that better-published evaluators influence decisions both through their own assessments and through their influence on the criteria and effort of their fellow evaluators. These peer responses offer one of the possible explanations for the gains in selection quality, although we cannot quantify their contribution. The increase in measured effort is consistent with the reputational mechanism in Swank and Visser (2023): evaluators may work harder out of concern for their reputation among more accomplished colleagues. The convergence in evaluation criteria may also arise as members share information, revise their views in response to colleagues' arguments, or defer to the judgments of more accomplished researchers.

4.3 Committee composition and nonlinear returns to expertise

Our baseline specification measures committee quality as the mean productivity of its members. The presence of peer influence may instead suggest that the first top researcher is especially consequential. Even a single top researcher may be able to identify information that would otherwise be missed, persuade other members, or affect the criteria against which all candidates are assessed. Conversely, if more accomplished experts provide distinct information or reinforce one another in deliberation, their contributions could instead be complementary. Individual evaluative ability may also rise nonlinearly with research accomplishment: gains in evaluative ability may flatten above a sufficient level of research accomplishment or become larger near the research frontier.

We explore these possibilities by distinguishing committees with one, two, or at least three top researchers, using the top 25%, 10%, 5%, and 1% of the field-specific productivity distribution as alternative thresholds to define top researchers. Appendix Table A13 reports subsequent citations and full-professor promotion among qualified and rejected associate-professor applicants.

The estimates provide some suggestive evidence that the gains from committee expertise are concentrated toward the top of the research-quality distribution and that the first highly qualified member may be particularly consequential for selecting candidates with high promotion potential. A single researcher in the top 5% or 1% can improve the citation outcomes of qualified candidates. Promotion outcomes suggest smaller incremental gains beyond the first top researcher, whereas citation outcomes show no consistent pattern of diminishing returns. However, the estimates are too imprecise, hence we retain average committee quality as our baseline measure.

5 Opportunity Costs

In this section, we exploit random assignment of evaluators to estimate the effects of committee service on research output, PhD supervision, and future volunteering. We first examine average research losses and how they vary by discipline, workload, team size, and estimate reductions in student supervision. Then we estimate whether costs vary with researchers' baseline productivity and whether these differences translate into differences to volunteer for future service.

Foregone research output. Using all potential evaluators who could have been assigned to ASN committees between 2012 and 2021, we estimate:

$$y_{j,e,t+k} = \beta_0 + \beta_1 D_{j,e} + \mathbf{X}_j'\mathbf{\Gamma} + \mu_e + \varepsilon_{j,e,t+k} \tag{4}$$

where $y_{j,e,t+k}$ denotes the research output of researcher j, k years following the time of evaluator assignment to examination e and $D_{j,e}$ indicates whether the researcher served on the committee.[26] We standardize productivity measures to have mean zero and standard deviation one within each examination's pool of potential evaluators. To improve precision, we control for researchers' pre-assignment publication counts, the share of high-impact articles, and a quadratic in academic age. We also include examination fixed effects, μ_e.

Following the recentering approach described in Section 4, we instrument actual service $D_{j,e}$ with the indicator for initial selection minus the researcher's assignment probability, estimated by simulating the committee draw. The corresponding balance check for potential evaluators shows no systematic association between this instrument and their predetermined characteristics (Appendix Table A14). Few initially selected evaluators resign, so initial selection strongly predicts actual service (Appendix Table A15). Under the standard IV assumptions, the coefficient β_1 identifies the causal effect of committee service on subsequent research output for researchers whose participation is induced by initial draw.[27] We estimate Equation (4) by 2SLS and cluster standard errors at the examination level.

Panel A of Figure 1 shows that committee service reduces annual publications by approximately

26. The estimation sample varies across time horizons because research output is observed only through 2024. For example, evaluators assigned in 2021 are observed for up to three years after assignment and are therefore excluded from the estimation sample at longer horizons.

27. The exclusion restriction states that the initial assignment may only affect researchers through committee service. Monotonicity requires that there are no evaluators who would only serve if they were drawn as replacements and resign if they were initially allocated. Neither of these assumptions is likely to be violated in our setting.

0.05 standard deviations in each of the first three years following the draw. From the fourth year onward, the estimates are close to zero, with no clear evidence of a compensating increase in output. Panel B reports effects on cumulative publications, measured from the year of assignment onward and set to zero in the preceding year.[28] Within the four years following the evaluator assignment (event time years 0–3), committee service reduces the total number of accumulated publications by approximately 0.05 standard deviations. This corresponds to roughly 1.9 publications, or 28% of an average evaluator's annual output.[29] The cumulative point estimates remain negative even ten years following the initial assignment, suggesting that the initial publication losses may not be fully recovered, however, the longer-run estimates are imprecise and cannot rule out a full catch-up.

These estimates likely provide a lower bound on the cost of service among all researchers: they apply to researchers who volunteer and whose participation responds to random assignment. Researchers expecting especially high costs may not volunteer and may also be more likely to resign. Both forms of selection would introduce upward bias, pushing the estimated loss closer to zero.

Heterogeneity by field, evaluation workload, and team size. Table 6 shows that committee service has a small and statistically insignificant effect on publication output in STEM+M, but reduces output in SSH by approximately 0.1 standard deviations over the four years following committee assignment. To explore potential explanations, we split the sample within each broad field by expected evaluation workload and collaboration team size. We proxy workload using the discipline-level average total number of words written per evaluator across individual reports in 2012, capturing both the number of applications and the detail of their assessments. Team size is the researcher's average number of authors per paper during the ten years preceding the examination.

We find no statistically significant differences in the effects of service across workload groups within either STEM+M or SSH. By contrast, the decline in SSH publication output is concentrated among researchers working in smaller teams. Effects in STEM+M are smaller and show no clear team-size gradient. Comparisons across fields are complicated by limited overlap in team-size distributions: even SSH researchers in the larger-team group tend to work in smaller teams than STEM+M researchers in the smaller-team group.

28. For pre-assignment years, the outcome is defined symmetrically as a sum of publications between the year in question and the year preceding assignment.

29. Cumulative output over event time periods 0–3 has a standard deviation of 39 publications and a mean of 28, equivalent to roughly seven publications per year (Appendix Table A2, Panel B).

These results are consistent with larger teams being better able to redistribute research tasks when a member undertakes committee service. However, smaller publication losses among researchers in larger teams may also reflect differences in the types of projects they undertake or the resources available to them. If collaborators sustain output by working more, our measures would not directly capture that additional effort. Smaller losses in evaluators' publication output therefore need not imply lower overall costs of committee service.

To account for team-size heterogeneity when later examining costs by evaluator productivity, we allow the effect of service to vary continuously with inverse baseline team size:

$$y_{j,e,t+k} = \beta_0 + \beta_1 \frac{D_{j,e}}{\bar{\alpha}_{j,e}} + \beta_2 \frac{1}{\bar{\alpha}_{j,e}} + \mathbf{X}_j' \mathbf{\Gamma} + \mu_e + \varepsilon_{j,e,t+k} \tag{5}$$

where $\bar{\alpha}_{j,e}$ denotes the average number of authors per paper of researcher j during the ten years preceding the draw for exam e. We instrument the scaled treatment variable, $D_{j,e}/\bar{\alpha}_{j,e}$, with the correspondingly scaled initial random draw, $\widetilde{Z}_{j,e}/\bar{\alpha}_{j,e}$.

Under this specification, β_1 is the effect of service for a researcher whose baseline publications are all sole-authored. For other researchers, the effect of service equals to $\beta_1/\bar{\alpha}_{j,e}$: its magnitude declines in inverse proportion to team size. Modeling this variation with a single treatment coefficient provides a parsimonious way to account for team size and can improve precision in the subsequent analysis of costs by evaluator productivity. A specification including both the unscaled treatment indicator and the treatment scaled by inverse team size favors the scaled specification (Appendix Table A16).[30] We interpret this relationship as heterogeneity in the effects of committee service, rather than as the causal effect of collaborators.

Panel A of Figure 2 reports the cumulative estimates on research output. For a sole author, committee service reduces output over the subsequent four years by approximately 0.19 standard deviations, with statistically significant effects lasting up to eight years. However, sole authors constitute less than 5% of the sample. At the sample mean of 6 authors per paper, the loss within the first four years is approximately 0.03 standard deviations, or 1.2 publications.

Foregone student supervision. We next examine whether committee service affects PhD supervision. We estimate Equation (5) with the number of supervised PhD graduates as the outcome, restricting the sample to universities that systematically report supervision activity.

30. The results are robust to scaling by the inverse square and inverse square root of team size.

Panel B of Figure 2 shows that committee service reduces PhD supervision. The effects become statistically significant five years after the assignment of evaluators, consistent with reduced intake of new PhD students rather than delayed completion of ongoing supervision. In the long run, an evaluator with an average number of coauthors who serves on a committee supervises approximately 0.4 fewer PhD graduates.

Heterogeneity by evaluator quality. We extend Equation (5) by adding an interaction of the scaled treatment status with evaluators' baseline quality. We instrument this interaction using the scaled assignment instrument interacted with the same quality measure. Table 7 reports estimated effects over three- and eight-year horizons for researchers with average baseline quality and those with one standard deviation above the mean. All effects are evaluated at the sample mean baseline team size of 6 authors per paper.[31]

Better-published researchers experience larger losses in research output. Over four years, committee service reduces total output by 0.032 standard deviations for a researcher of average quality and by 0.055 standard deviations for a researcher whose baseline quality is one standard deviation above the mean. This represents a loss that is approximately 1.7 times larger and this difference is statistically significant ($p = 0.018$). The losses also represent a larger share of better-published researchers' usual output: researchers one standard deviation above the mean are roughly 1.5 times more productive than the average researcher. The pattern extends across publication categories, although the difference for high-impact articles is not statistically significant.

In the long-run, eight years after the committee assignment, the point estimates imply a loss in total output of 0.015 standard deviations for a researcher of average quality and 0.035 standard deviations for a researcher one standard deviation above the mean. This loss is approximately 2.3 times as large. While this difference is economically substantial, it is imprecisely estimated ($p = 0.101$).

We find no statistically significant differences in supervision losses by researcher quality (Column 8 of Table 7). Consistently, descriptive statistics show that the number of PhD graduates supervised is broadly similar across researchers of different productivity levels based on research output (Appendix Table A2).

31. We rescale the cost to more productive and average researchers by the same factor, since researchers of different baseline research quality have similar numbers of authors per paper (Appendix Table A2). Under this specification, the proportional difference between researchers of different baseline productivity is unchanged when both effects are rescaled to any other common team size.

Future participation of selected researchers. Finally, we examine whether the costs of committee service affect researchers' willingness to volunteer again to serve on an evaluation committee. Aggregate trends show that top researchers were initially more likely to volunteer, but their participation declined over time (Appendix Figure A1).

We estimate Equation (5) using an indicator for volunteering in the next cycle in which all members of the relevant evaluator pool are legally eligible as the outcome variable. For the 2012 cohort, this is the 2018 cycle; for the 2016 cohort, it is 2023. We exclude exams in 2018 and 2021 from this analysis because we do not observe a subsequent cycle in which previously selected evaluators would be eligible.

A challenge in this analysis is that participation rules create a mechanical relationship between past assignment and later eligibility. Evaluators not drawn in the initial cycle may serve in an intermediate cycle and thereby become ineligible for the later cycle used as the outcome.

Column 1 of Table 8 reports estimates from the unrestricted sample, which conflate behavioral responses with with temporary ineligibility following committee service. To remove this mechanical effect, we exclude evaluators who were not initially drawn but subsequently served in an intermediate cycle – 2016 for the 2012 cohort and 2018–2021 for the 2016 cohort. This restriction may induce some selection because researchers more willing to volunteer are more likely to serve in intermediate cycles, but it eliminates the direct eligibility distortion. On the restricted sample we observe that prior service significantly reduces subsequent volunteering, especially among more productive researchers (Columns 2 and 3). For a researcher with the average number of collaborators, assignment to committee service lowers the probability of volunteering again by 1.4 percentage points. For a researcher with the same number of collaborators but who is one standard deviation better-published, this effect is almost twice as large at a 2.6 percentage point decrease. This difference is statistically significant ($p = 0.044$).

Committee service thus carries costs beyond evaluators' own research, reducing PhD supervision and discouraging future participation. Larger research losses and declines in volunteering among more productive researchers suggest that sustaining their involvement may be particularly challenging.[32]

32. We find no evidence that the effects of committee service on research output or subsequent volunteering vary with fellow evaluators' research quality (Table A17).

6 Cost-benefit Calculation and the Research Output–Time Elasticity

We put our findings in perspective through two complementary exercises. First, we provide a rough comparison of the costs and benefits of involving better-published evaluators. Second, we provide an approximation of the elasticity of research output with respect to researchers' time. Both rely on additional assumptions and are specific to the Italian setting, but offer a sense of the magnitudes involved.

6.1 Back-of-the-envelope cost-benefit calculation

The benefits and costs of evaluation service are difficult to compare directly: benefits materialize in candidates' subsequent research impact and career progression, while costs appear in evaluators' own output and PhD supervision. Future evaluator supply provides one margin on which the benefits and costs of involving accomplished researchers can be compared. Better-published committees select candidates who are more likely to enter future evaluator pools as top researchers, while service reduces evaluators' own willingness to participate again. We examine whether this replenishment could offset the loss of potential evaluators.

Consider replacing an average committee member with a researcher whose publication record is one standard deviation stronger counting benefits only among the ≈ 258 associate-professor applicants per committee. An average committee would qualify approximately 95.5 candidates at the baseline qualification rate of 37%. Our estimates in Tables 1 and 3 imply that the replacement reduces this number to approximately 92, but raises the share of qualified candidates who subsequently enter the evaluator pool as top researchers from 4% to 4.275%. The expected number of future top evaluators among qualified candidates therefore rises from approximately 3.82 to 3.93. The resulting gain of 0.11 accounts for both the stronger composition of the selected group and the smaller number of qualified candidates.[33]

On the cost side, estimates in Table 8 show that committee service reduces subsequent volunteering by approximately 2.6 percentage points for a researcher whose baseline productivity is one standard deviation above the mean and whose team size equals the sample average of six authors. This corresponds to an expected loss of 0.026 potential evaluators. Under the more demanding scenario in which the researcher works alone potentially internalizing the largest share of service

33. Replacing one of the four domestic committee members increases average committee research productivity by one quarter of a standard deviation. The calculation is therefore $258[(0.37-0.054/4)(0.04+0.011/4)-0.37\times 0.04] \approx 0.11$. For this illustrative exercise, we assume that rejected candidates do not subsequently replenish the evaluator pool.

costs, scaling this loss by team size gives approximately 0.16.

The estimated replenishment of 0.11 future top evaluators thus exceeds the participation loss at the average team size, but falls short of the loss in the sole-author scenario. These calculations suggest that better selection may not fully offset reduced participation, underscoring the importance of retaining accomplished researchers in evaluation roles.

6.2 The elasticity of research output with respect to time

Committee service reduces cumulative publications from the assignment year through year three by the equivalent of 28% of a typical year's output. To relate this loss to research time, we consider an illustrative workload of 200–300 hours over the two-year mandate.[34] If committee work displaced research time hour for hour, this would represent 19–29% of an annual research-time benchmark of approximately 1,040 hours (Bentley and Kyvik, 2012), implying an output–time elasticity of approximately 1.0–1.5. Applying the same assumptions on the workload to the larger publication loss in SSH, equivalent to 45% of annual output, yields an elasticity of rouughly 1.6–2.4.

Teaching relief could reduce the research time displaced and therefore increase the implied elasticity. University records document teaching reductions associated with ASN service, including allowances between 40 and 100 hours for an academic year. However, relief had to be accommodated without additional public expenditure, and the records also document evaluators who did not request it or whose requests were refused.[35]

If workload adjustments preserved 100 research hours over the mandate, the implied elasticity would rise to approximately 1.5–2.9. These calculations remain illustrative: we do not observe evaluation hours nor the extent to which teaching relief preserved research time directly.

7 Conclusion

In this paper, we use a large-scale randomized natural experiment in Italian academia to study the benefits and costs of involving accomplished researchers in scientific evaluation. Better-published evaluators make more informative decisions: their qualification decisions better predict candidates' subsequent research performance. Committee service, however, reduces evaluators' research output, PhD supervision, and subsequent volunteering.

34. Informal conversations with economics evaluators suggest a time commitment in this range.

35. Informal accounts likewise suggest that many evaluators did not request teaching relief. Appendix B.3 describes the documentary evidence, including cases of approved reductions, non-take-up, and refusal.

The trade-off is particularly pronounced for better-published researchers, who incur larger research losses and larger declines in willingness to serve again. Research losses are also larger in the Social Sciences and Humanities, particularly among researchers working in smaller teams. The benefits of expertise likewise vary across decisions. In our setting, they are concentrated among associate-professor applicants, consistent with expertise being more valuable when research records are less established.

These findings suggest possible responses by institutions that commission evaluations. Funding agencies, publishers, and academic evaluation bodies could consider which decisions benefit most from accomplished researchers' expertise, while paying particular attention to the research costs of service in fields where researchers typically work alone or in small teams. They could also consider compensating evaluators for their time and funding workload relief or research support. Such support may help sustain participation, since service reduces researchers' willingness to evaluate again. Universities could complement these measures by accounting for substantial evaluation duties in workload allocation and career assessments.

For individual researchers, our findings offer an additional consideration when deciding which evaluation duties to accept. Reduced PhD supervision suggests that service may affect the training of future researchers, while smaller publication losses in larger teams are consistent with collaborators absorbing part of the disruption. These potential costs, alongside those to one's own research, can be weighed against the contribution that one's expertise is likely to make in the particular evaluation under consideration.[36]

A similar trade-off may arise wherever institutions draw on practicing experts, including hiring panels, professional licensing boards, and regulatory commissions. Their judgment can improve decisions, but providing it takes time away from other valuable work. Where participation is voluntary, these opportunity costs may also affect who remains willing to serve.

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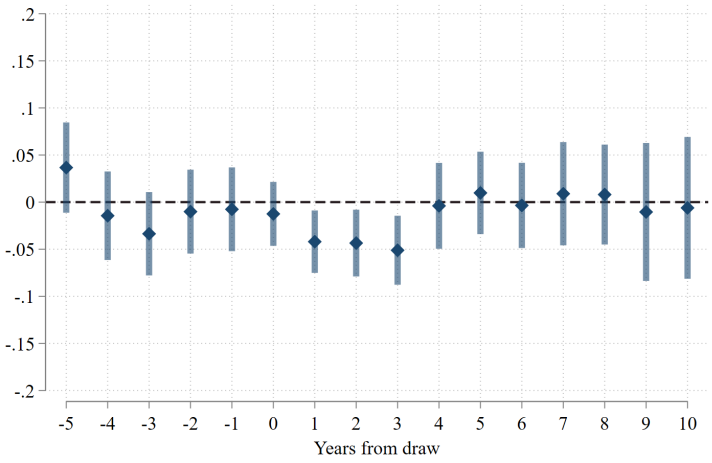
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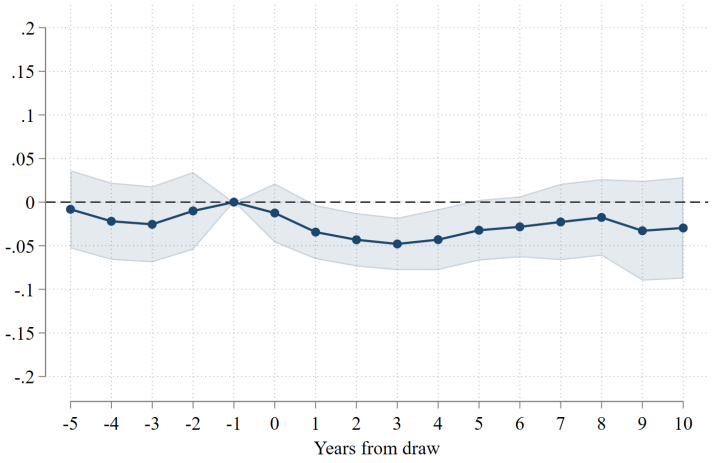
Figures

Figure 1: Effect of committee service on research output

Panel A.
Annual Research Output



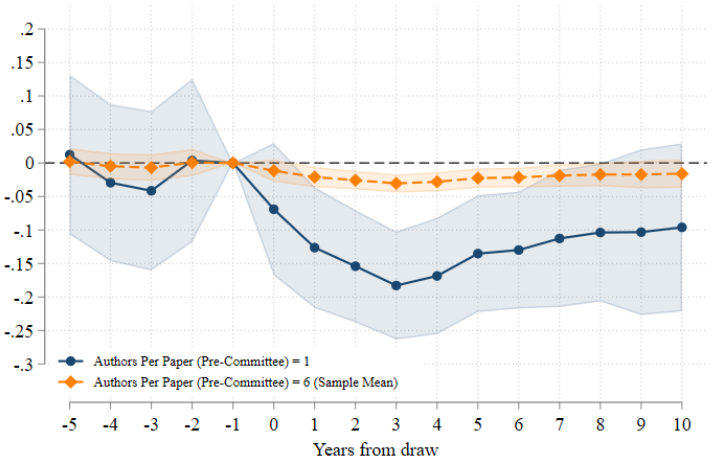
Panel B.
Cumulative Research Output



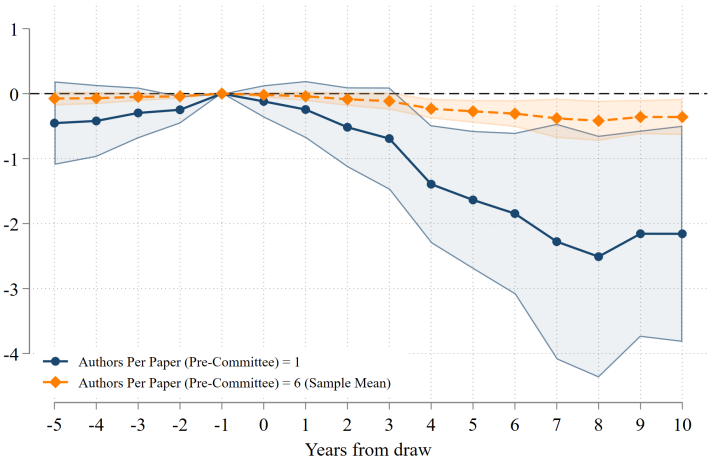
Notes: The sample includes potential ASN evaluators in the 2012–2021 cycles. Estimates at longer horizons use only earlier cycles, with available information on long-term outcomes. Estimates are based on Equation (4). Panel A reports annual research output. Panel B reports cumulative output relative to the year before assignment, with accumulation defined symmetrically before and after that year. Both outcomes are standardized within each examination's evaluator pool. All specifications include examination fixed effects; post-assignment estimates also control for pre-assignment research output and a quadratic in academic age. Standard errors are clustered at the exam level. Vertical bars and shading indicate 95% confidence intervals.

Figure 2: Effect of committee service on research output and PhD supervision by number of authors per paper

Panel A. Cumulative Research Output



Panel B. Cumulative PhD Graduates Supervised



Notes: The sample includes potential ASN evaluators in the 2012–2021 cycles. Panel B is restricted to researchers at universities with consistent PhD reporting over the sample period. Estimates at longer horizons use only earlier cycles with sufficient follow-up. Estimates are based on Equation (5). Outcomes are cumulative research output (Panel A) and the cumulative number of PhD graduates supervised (Panel B), accumulated relative to the year before assignment. Research output is standardized within each examination's evaluator pool. The solid blue line shows estimated effects for sole authors; the dashed orange line shows effects at the sample's average team size of six authors per paper. Team size is measured over the ten years before assignment. All specifications control for examination fixed effects and the inverse number of authors per paper. Post-assignment estimates also control for pre-assignment research output and a quadratic in academic age. Standard errors are clustered at the exam level. Shaded areas indicate 95% confidence intervals.

Tables

Table 1: Committee composition and evaluation outcomes

	Qualified				Promoted			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
			STEMM	SSH			STEMM	SSH
Candidate's Number of Publications	0.121***	0.121***	0.127***	0.110***	0.087***	0.087***	0.090***	0.083***
	(0.004)	(0.004)	(0.006)	(0.005)	(0.003)	(0.003)	(0.004)	(0.004)
Candidate's Impact of Publications	0.093***	0.093***	0.088***	0.100***	0.039***	0.039***	0.031***	0.051***
	(0.006)	(0.005)	(0.007)	(0.008)	(0.003)	(0.002)	(0.003)	(0.004)
Committee Quality	-0.054*	-0.054*	-0.058*	-0.042	0.012	0.012	0.017	0.001
	(0.029)	(0.029)	(0.029)	(0.065)	(0.014)	(0.014)	(0.016)	(0.024)
Candidate's Number of Publications × Committee Quality		-0.026**	-0.026	-0.019		-0.007	-0.010	0.002
		(0.012)	(0.017)	(0.016)		(0.007)	(0.009)	(0.013)
Candidate's Impact of Publications × Committee Quality		0.035***	0.025*	0.052***		0.020***	0.015*	0.023**
		(0.012)	(0.015)	(0.018)		(0.007)	(0.009)	(0.011)
Observations	69020	69020	41395	27625	69020	69020	41395	27625
Adjusted R^2	0.093	0.094	0.098	0.092	0.052	0.052	0.054	0.052
Exams	184	184	109	75	184	184	109	75

Notes: The sample includes applications submitted in the first wave of the 2012 ASN cycle. Outcomes are indicators for qualification (Columns 1–4) and promotion to the rank sought within five years (Columns 5–8). Candidate productivity measures are standardized within examinations; evaluator quality is standardized among all full professors at Italian public universities within discipline-year cells. All specifications include four broad-disciplinary-group indicators and controls for evaluator–candidate connections. Standard errors are clustered at the exam level.
*** p < .01, ** p < .05, * p < .1

Table 2: Committee composition and subsequent outcomes of qualified and rejected candidates

	Future Research Outcomes				
	(1) Total Works	(2) High-Impact Articles	(3) Total Citations	(4) Citations to Pre-Exam Works	(5) Citations to Post-Exam Works
Panel A. Candidates to Associate and Full Professor					
Qualified candidates					
Committee quality	0.003	0.008	0.043***	0.027**	0.044***
	(0.020)	(0.019)	(0.014)	(0.012)	(0.015)
Non-qualified candidates					
Committee quality	-0.005	-0.003	-0.018*	-0.009	-0.018*
	(0.014)	(0.017)	(0.009)	(0.008)	(0.009)
Baseline mean: qualified	0.24	0.26	0.15	0.18	0.11
Baseline mean: non-qualified	-0.14	-0.15	-0.09	-0.11	-0.06
Panel B. Candidates to Associate Professor					
Qualified candidates					
Committee quality	0.017	0.023	0.058***	0.039***	0.065***
	(0.026)	(0.023)	(0.019)	(0.014)	(0.021)
Non-qualified candidates					
Committee quality	-0.005	-0.001	-0.026**	-0.014	-0.030**
	(0.016)	(0.017)	(0.012)	(0.010)	(0.012)
Baseline mean: qualified	0.26	0.27	0.15	0.18	0.11
Baseline mean: non-qualified	-0.15	-0.16	-0.09	-0.11	-0.06
Panel C. Candidates to Full Professor					
Qualified candidates					
Committee quality	-0.027	-0.025	0.006	-0.000	-0.004
	(0.021)	(0.028)	(0.019)	(0.014)	(0.024)
Non-qualified candidates					
Committee quality	0.001	-0.001	0.001	0.005	0.010
	(0.017)	(0.024)	(0.012)	(0.009)	(0.015)
Baseline mean: qualified	0.20	0.24	0.15	0.19	0.10
Baseline mean: non-qualified	-0.11	-0.13	-0.08	-0.10	-0.06

Notes: The sample includes applicants in the first wave of the 2012 ASN cycle. The estimates are based on Equation (2), separately for qualified and rejected candidates. Panel A pools applications to Associate and full professor. Panels B and C restrict the sample to associate professor and full professor applications, respectively. Future research performance is measured over the ten years following the examination and is standardized within exam. All specifications control for candidates' pre-evaluation publication quantity, publication impact, and citations; broad disciplinary group, institution, and academic rank; and evaluator–candidate connections. Standard errors are clustered at the exam level.
*** p < .01, ** p < .05, * p < .1

Table 3: Committee composition and subsequent career outcomes of qualified and rejected associate professor candidates

	Future Career Outcomes		
	(1) Promoted to Full Professor	(2) Future Evaluator	(3) Future Top Evaluator
Qualified candidates			
Committee quality	0.038***	0.029***	0.011**
	(0.014)	(0.009)	(0.004)
Non-qualified candidates			
Committee quality	0.006	0.000	0.000
	(0.004)	(0.003)	(0.002)
Baseline mean: qualified	0.17	0.10	0.04
Baseline mean: non-qualified	0.04	0.02	0.01

Notes: The sample includes associate professor applicants in the first wave of the 2012 ASN cycle. The estimates are based on Equation (2), separately for qualified and rejected candidates. Columns 1–3 indicate whether candidates became full professors, entered a subsequent evaluator pool, or entered it as a top researcher (top 25% by high-impact publication count among Italian full professors in their field and year) within ten years. All specifications control for candidates' pre-evaluation publication quantity, publication impact, and citations; broad disciplinary group, institution, and academic rank; and evaluator–candidate connections. Standard errors are clustered at the exam level.
*** p < .01, ** p < .05, * p < .1

Table 4: Committee composition and individual voting

	Positive Vote			
	(1)	(2)	(3)	(4)
Evaluator Quality	-0.023**	-0.000	-0.023**	-0.023**
	(0.011)	(0.002)	(0.011)	(0.011)
Peer Quality	-0.055**		-0.055**	-0.055**
	(0.024)		(0.024)	(0.024)
Evaluator Quality × Candidate's Number of Publications			-0.005**	-0.005
			(0.003)	(0.004)
Evaluators Quality × Candidate's Impact of Publications			0.006***	0.007**
			(0.002)	(0.004)
Peer Quality × Candidate's Number of Publications				-0.017*
				(0.010)
Peer Quality × Candidate's Impact of Publications				0.030***
				(0.009)
Observations	241744	241744	241744	241744
Candidate Fixed Effects		✓		

Notes: The sample includes individual evaluator reports from the first wave of the 2012 ASN cycle. The estimates are based on Equation (3). The outcome indicates whether the evaluator voted to qualify the candidate. *Peer quality* is the average research productivity of the other committee members, measured by their standardized number of high-impact publications in the ten years before evaluation. Controls include expected peer quality, the evaluator's (re-centered) probability of being drawn, candidate productivity, and evaluator–candidate connections. Standard errors are clustered at the exam level.
*** p < .01, ** p < .05, * p < .1

Table 5: Committee composition and individual report writing

	Words	Average IDF	Total IDF
	(1)	(2)	(3)
Evaluator Quality	-0.029	-0.001	-0.017
	(0.042)	(0.036)	(0.039)
Peer Quality	0.203**	0.171**	0.199**
	(0.097)	(0.085)	(0.095)
Observations	241744	241744	241744

Notes: The sample includes individual evaluator reports from the first wave of the 2012 ASN cycle. The estimates are based on Equation (3). Outcome variables are textual features of reports, standardized within macro-field and application-rank cells. *Peer quality* is the average quality-adjusted publication record of evaluators, excluding j, over the ten years preceding the exam. Controls include the expected quality of peers, researchers' (re-centered) probability of being drawn, candidate productivity, and evaluator–candidate connections. Standard errors are clustered at the exam level.
*** p < .01, ** p < .05, * p < .1

Table 6: Effects of committee service on research output by discipline, evaluation workload, and team size

Panel A: STEMM					
	Overall	Evaluation Workload		Collaboration Team Size	
	(1)	(2)	(3)	(4)	(5)
		Low	High	Low	High
Committee Service	-0.024	-0.020	-0.038	-0.031	-0.018
	(0.018)	(0.027)	(0.026)	(0.029)	(0.025)
Observations	9775	4349	5248	4878	4834
Average Number of Authors per Paper	7.96	7.43	8.33	4.37	11.60
Average Number of Words	59131	23573	88597	63083	55229
Panel B: SSH					
	Overall	Evaluation Workload		Collaboration Team Size	
	(1)	(2)	(3)	(4)	(5)
		Low	High	Low	High
Committee Service	-0.101***	-0.110***	-0.080**	-0.181***	-0.045
	(0.027)	(0.040)	(0.037)	(0.040)	(0.041)
Observations	5916	2611	3281	2921	2988
Average Number of Authors per Paper	2.56	2.48	2.63	1.36	3.74
Average Number of Words	77537	29468	115791	84359	71006

Notes: The sample comprises all potential evaluators in the ASN system during 2012–2021. The estimates are based on Equation (4). The outcome is cumulative research output over event years 0–3, standardized within examination pools. Panels A and B report results for STEM+M and SSH, respectively. Column (1) uses the full sample within each panel. Columns (2)–(3) split disciplines by median evaluation workload, measured by the average total words written per evaluator in the corresponding field in 2012. Columns (4)–(5) split researchers by median team size, measured by average authors per paper over the ten years preceding assignment. Both splits are calculated separately within STEM+M and SSH. All specifications include examination fixed effects, pre-assignment research output, and a quadratic of academic age. Standard errors are clustered at the exam level.

*** $p < .01$, ** $p < .05$, * $p < .1$

Table 7: Effects of committee service on research output and PhD supervision by baseline research quality

	3 years post-exam				8 years post-exam			
	Total Works	High-impact Articles	Lower-impact Articles	Students Supervised	Total Works	High-impact Articles	Lower-impact Articles	Students Supervised
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Researcher at Mean Baseline Quality	-0.032***	-0.024***	-0.028***	-0.099	-0.015*	-0.010	-0.020*	-0.391**
	(0.008)	(0.008)	(0.009)	(0.068)	(0.009)	(0.010)	(0.011)	(0.159)
Researcher One Standard Deviation Above the Mean Baseline Quality	-0.055***	-0.039**	-0.049***	-0.137	-0.035**	-0.019	-0.039*	-0.561**
	(0.013)	(0.016)	(0.017)	(0.134)	(0.017)	(0.022)	(0.021)	(0.245)
p-value: Equality of Effects	0.018	0.181	0.086	0.729	0.101	0.524	0.212	0.382
Observations	15,712	15,712	15,712	7,224	9,552	9,552	9,552	3,982
Number of Exams	630	630	630	628	347	347	347	345

Notes: The sample includes all potential ASN evaluators in 2012–2021; estimates at longer horizons use earlier cycles with observed outcomes. Estimates follow Equation (5), allowing the effect of committee service to vary with baseline research quality. Effects are reported at mean quality and one standard deviation above the mean, evaluated at the sample average of six authors per paper. Research output is accumulated from the year prior to the exam, normalized within each examination's pool of potential evaluators. PhD supervision is measured in levels and restricted to institutions that consistently reported graduates. The reported p-value tests equality of the effects at the two quality levels. All specifications control for examination fixed effects, inverse team size, past research output, and a quadratic in academic age. Standard errors are clustered at the exam level.
*** p < .01, ** p < .05, * p < .1

Table 8: Effects of committee service on subsequent volunteering

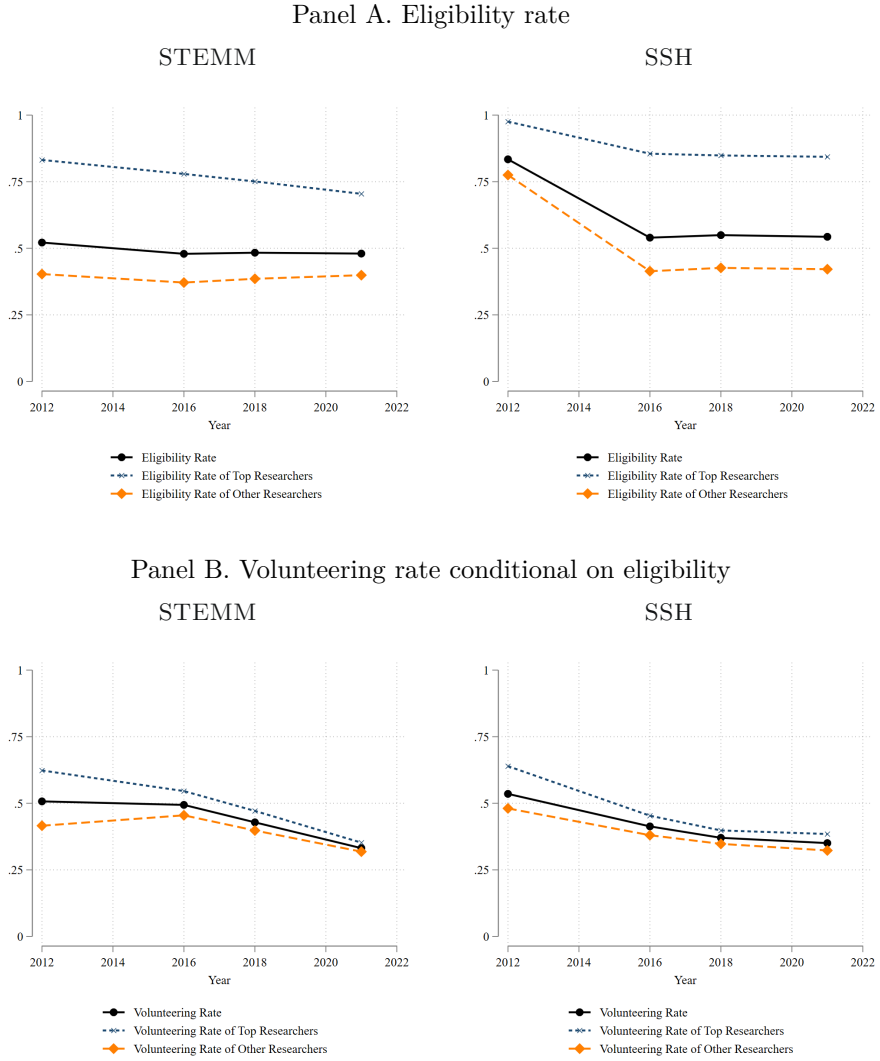
	Volunteer Again		
	Unrestricted Sample	Restricted Sample	Restricted Sample
	(1)	(2)	(3)
Pooled Effect	-0.001	-0.014**	
	(0.005)	(0.005)	
Researcher at Mean Baseline Quality			-0.014***
			(0.005)
Researcher One Standard Deviation Above the Mean Baseline Quality			-0.026***
			(0.008)
p-value: Equality of Effects			0.044
Observations	9,552	8,696	8,696
Number of Exams	347	347	347
Restricted Sample	No	Yes	Yes
Exam Fixed Effects	Yes	Yes	Yes

Notes: The sample includes potential ASN evaluators in 2012 and 2016. The outcome indicates volunteering in 2018 for the 2012 cohort and in 2023 for the 2016 cohort. Columns (2)–(3) exclude researchers drawn for service in intervening cycles, which would make them ineligible to volunteer. The estimates are based on Equation (5). Columns (1)–(2) report pooled effects; Column (3) allows effects to vary with baseline research quality and reports estimates at the mean and one standard deviation above it. All effects are evaluated at the sample average of six authors per paper. The reported p-value tests equality of the effects at the two quality levels. All specifications control for examination fixed effects, inverse team size, past research output, and a quadratic in academic age. Standard errors are clustered at the exam level.
*** p < .01, ** p < .05, * p < .1

APPENDIX FOR ONLINE PUBLICATION

A Appendix Figures and Tables

FIGURE A1: EVALUATOR ELIGIBILITY AND VOLUNTEERING RATES



Notes: The sample includes full professors at Italian public universities in 2012–2021. Panel A reports the share eligible to serve as evaluators, with eligibility simulated using the Ministry's official criteria. Panel B reports the share volunteering among those eligible. *Top researchers* are in the top quartile of quality-adjusted research output within their field, measured among Italian full professors over the ten years preceding each evaluation cycle. The solid black line represents all professors, the short-dashed blue line top researchers, and the long-dashed orange line other researchers.

Table A1: Evaluator characteristics by eligibility, volunteering, and resignation status

Panel A. Eligibility

	Not Eligible		Eligible		Difference	
	Mean	SD	Mean	SD	Diff	P-value
Top Researcher	0.11	0.32	0.42	0.49	-0.30***	0.00
Total Works	-0.36	0.73	0.38	1.05	-0.73***	0.00
Articles	-0.36	0.73	0.37	1.05	-0.73***	0.00
High-Impact Publications	-0.37	0.69	0.37	1.08	-0.74***	0.00
Lower-Impact Publications	-0.25	0.79	0.28	1.08	-0.53***	0.00
Observations	23859		28544		52403	

Panel B. Volunteering conditional on eligibility

	Not Volunteered		Volunteered		Difference	
	Mean	SD	Mean	SD	Diff	P-value
Top Researcher	0.38	0.48	0.47	0.50	-0.09***	0.00
Total Works	0.27	1.03	0.52	1.06	-0.25***	0.00
Articles	0.26	1.04	0.51	1.06	-0.25***	0.00
High-Impact Publications	0.28	1.06	0.48	1.10	-0.20***	0.00
Lower-Impact Publications	0.18	1.05	0.40	1.10	-0.22***	0.00
Observations	16100		12444		28544	

Panel C. Resignation conditional on being drawn

	Not Resigned		Resigned		Difference	
	Mean	SD	Mean	SD	Diff	P-value
Top Researcher	0.40	0.49	0.45	0.50	-0.05*	0.08
Total Works	0.30	1.02	0.41	1.04	-0.11*	0.07
Articles	0.30	1.01	0.40	1.04	-0.10*	0.10
High-Impact Publications	0.28	1.05	0.41	1.17	-0.13*	0.06
Lower-Impact Publications	0.25	1.04	0.30	0.99	-0.05	0.39
Observations	3085		331		3416	

Notes: Panel A includes full professors at Italian public universities in the year preceding each evaluation cycle (2012, 2016, 2018, and 2021). Panel B restricts this sample to professors whose simulated eligibility meets the Ministry's official criteria; Panel C includes professors initially drawn for committee service. Research output is measured over the ten years preceding each cycle and standardized within examinations. *High-impact* articles are published in top-quartile Web of Science journals ranked by AIS in STEM+M or A-list journals in SSH; all other articles are classified as *lower-impact*. *Top researchers* are in the top quartile of quality-adjusted research output within their field, measured among Italian full professors over the same ten-year period.
*** p < .01, ** p < .05, * p < .1

Table A2: Research output of potential evaluators

	Overall		Top Researchers		Other Researchers	
	Mean	SD	Mean	SD	Mean	SD
Panel A: Ten years before the exam						
Works	60	70	86	91	44	48
Articles	53	63	77	82	39	42
High-Impact Articles	22	34	38	46	13	18
Lower-Impact Articles	31	36	39	44	26	30
PhD Students Supervised	3	6	4	7	3	6
Number of Authors per Paper	6	5	6	5	6	5
Panel B: Years 0–3 (cumulative)						
Works	28	39	39	52	21	27
Articles	24	36	35	48	18	24
High-Impact Articles	9	18	15	25	6	10
Lower-Impact Articles	15	21	19	27	12	17
PhD Students Supervised	2	5	2	6	2	4
Panel C: Years 0–8 (cumulative)						
Works	50	79	72	102	35	54
Articles	45	74	65	95	31	51
High-Impact Articles	18	37	29	49	10	23
Lower-Impact Articles	27	42	36	52	21	31
PhD Students Supervised	4	8	4	9	3	7
Observations	16,683		6,325		10,358	
Number of Exams	685		685		685	
Number of Researchers	10,487		4,313		7,193	

Notes: The sample includes potential ASN evaluators in 2012–2021. The table reports means and standard deviations for all evaluators and separately for top and other researchers. Panel A covers the ten years preceding assignment; Panels B and C cover event years 0–3 and 0–8, respectively. PhD supervision is restricted to institutions that consistently reported graduates. *High-impact* articles are published in top-quartile Web of Science journals ranked by AIS in STEM+M or A-list journals in SSH; all other articles are classified as *lower-impact*. *Top researchers* are in the top quartile of quality-adjusted research output within their field, measured among Italian full professors over the ten years preceding assignment. Observations are at the researcher–exam level; the numbers of exams and researchers refer to distinct exams and researchers within each column. Researchers may appear in multiple exams and may change top-researcher status across exams.

Table A3: Effects of committee quality on application withdrawal

	Application Withdrawn (1)
Candidate's Number of Publications	-0.037***
	(0.002)
Candidate's Impact of Publications	-0.028***
	(0.003)
Committee Quality	0.013
	(0.012)
Candidate's Number of Publications × Committee Quality	-0.003
	(0.005)
Candidate's Impact of Publications × Committee Quality	-0.002
	(0.008)
Observations	69020
Number of Exams	184
R^2	0.02
Candidate Connections	✓

Notes: The sample includes applications submitted in the first wave of the 2012 ASN cycle. The outcome indicates withdrawal after the committee and evaluation criteria were announced. *Candidate's Number of Publications* measures total research output; *Candidate's Impact of Publications* is the share of articles in top-quartile Web of Science journals in STEM+M or A-list journals in SSH. Both measures are standardized within examinations. *Committee Quality* is members' average quality-adjusted research output over the ten years preceding the examination, with individual quality standardized among Italian full professors within each discipline. Realized committee quality is instrumented with initially drawn committee quality net of its expectation. All specifications include broad disciplinary group indicators and controls for evaluator–candidate connections. Standard errors clustered at the field level are in parentheses. Standard errors are clustered at the field level.
*** p < .01, ** p < .05, * p < .1

Table A4: Candidate characteristics in the 2012 evaluation cycle

	Overall		By Field				By Application Level			
			STEM		SSH		Associate		Full	
	Mean	SD	Mean	SD	Mean	SD	Mean	SD	Mean	SD
Year of First Publication	1997	8	1996	8	1998	8	1998	7	1992	8
Works	46	42	56	47	31	27	41	37	58	51
Articles	25	31	35	35	11	14	22	26	33	38
High-impact Articles	10	17	16	20	3	5	9	14	14	21
Observations	69020		41395		27625		47426		21594	
Number of Candidates	46329		27613		19388		34643		16747	

Notes: The sample includes candidates who applied in the first wave of the 2012 ASN cycle. Research output measures are constructed from candidates' curricula vitae. *High-impact* articles are published in top-quartile Web of Science journals ranked by AIS in STEM+M or A-list journals in SSH.

Table A5: Subsequent research output of candidates in the 2012 evaluation cycle

	Qualified Candidates				Rejected Candidates			
	All Fields		Economics		All Fields		Economics	
	Mean	SD	Mean	SD	Mean	SD	Mean	SD
Total Works	56	84	29	31	34	54	23	29
Articles	51	79	26	29	30	51	21	27
High-Impact Articles	21	41	9	10	11	24	6	11
Citations: Pre-Exam Works	1118	2693	418	678	549	1621	231	594
Citations: Post-Exam Works	1250	4786	434	1351	719	3617	382	2575
Observations	25342		2236		43678		3769	

Notes: The sample includes candidates who applied in the first wave of the 2012 ASN cycle. Research output is accumulated over ten years starting after the examination year. *High-impact* articles are published in top-quartile Web of Science journals ranked by AIS in STEM+M or A-list journals in SSH. Citations are counted over 2013–2022, separately for works published through 2012 (pre-exam) and from 2013 onward (post-exam).

Table A6: Descriptive statistics for individual evaluation reports

	Overall		STEM		SSH		Difference	
	Mean	SD	Mean	SD	Mean	SD	Diff	P-value
Positive Vote	0.45	0.50	0.46	0.50	0.43	0.49	0.03	0.000
Words	182.45	285.88	150.32	192.24	232.78	384.32	-82.46	0.000
Average IDF	2.54	0.76	2.29	0.54	2.94	0.87	-0.65	0.000
Total IDF	304.87	619.88	219.98	389.98	437.82	847.89	-217.84	0.000
Any Bibliometric Mention	0.45	0.50	0.47	0.50	0.43	0.50	0.04	0.000
Observations	241744		147538		94206		241744	
Number of Evaluators	758		450		308			
Number of Candidates	40217		23953		16769			

Notes: The sample includes all individual evaluator reports for applications submitted in the first wave of the 2012 ASN cycle. *Words* is the report's word count. *Average IDF* is the mean inverse document frequency score of its words. *Total IDF* equals *Words* × *Average IDF*. *Bibliometric indicators* indicates whether the report mentions bibliometric measures or databases, such as the h-index, citations, A-list articles, Scopus, or Web of Science.

Table A7: Report characteristics and candidate profiles

	Words			Average IDF			Total IDF		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Coauthors	0.123**	0.094***	0.115***	0.135***	0.106***	0.117***	0.148***	0.114***	0.134***
	(0.057)	(0.028)	(0.020)	(0.039)	(0.031)	(0.027)	(0.053)	(0.028)	(0.021)
Same University	0.109***	0.090**	0.089***	0.050	0.068**	0.072***	0.106***	0.094**	0.100***
	(0.039)	(0.037)	(0.019)	(0.035)	(0.034)	(0.024)	(0.038)	(0.037)	(0.019)
Same Subfield	0.169***	0.112***	0.128***	0.050*	0.115***	0.093***	0.148***	0.120***	0.138***
	(0.032)	(0.039)	(0.025)	(0.029)	(0.037)	(0.023)	(0.031)	(0.039)	(0.028)
Candidate's Number of Publications	0.057***			0.017*			0.058***		
	(0.009)			(0.009)			(0.009)		
Candidate's Impact of Publications	0.038***			0.029***			0.043***		
	(0.010)			(0.007)			(0.009)		
CVs to Evaluate (100s)	-0.008	-0.011		-0.023**	-0.051***		-0.014	-0.026	
	(0.019)	(0.027)		(0.012)	(0.018)		(0.018)	(0.025)	
Observations	241744	241744	241744	241744	241744	241744	241744	241744	241744
Candidate Fixed Effect		✓	✓		✓	✓		✓	✓
Evaluator Fixed Effect			✓			✓			✓

Notes: The sample includes all individual evaluator reports for applications submitted in the first wave of the 2012 ASN cycle. Outcomes measure textual features of reports. All measures are standardized within each combination of macro field (86 fields) and application level (associate or full professor). Standard errors are clustered at the exam level.
*** p < .01, ** p < .05, * p < .1

Table A8: Report characteristics and qualification outcomes

	Total Words			Average IDF			Total IDF		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Marginal Candidate	0.203***	0.167***	0.169***	0.007	-0.014	-0.026	0.191***	0.153***	0.161***
	(0.057)	(0.057)	(0.039)	(0.042)	(0.043)	(0.035)	(0.053)	(0.053)	(0.036)
Observations	241744	241744	241744	241744	241744	241744	241744	241744	241744
Candidate Controls		✓	✓		✓	✓		✓	✓
Evaluator Fixed Effect			✓			✓			✓

Notes: The sample includes all individual evaluator reports from the first wave of the 2012 ASN cycle. Outcomes measure textual features of reports. All measures are standardized within each combination of macro field (86 fields) and application level (associate or full professor). *Marginal Candidate* indicates whether the candidate received three or four favorable votes. *Candidate Controls* include publication counts and the share of high-impact articles. Standard errors are clustered at the exam level.
*** p < .01, ** p < .05, * p < .1

Table A9: Randomization check: candidate characteristics

	Publications	Share High-Impact	Citations
	(1)	(2)	(3)
Committee Quality	-4.960	-0.014	-79.163
	(3.753)	(0.024)	(90.716)
Observations	69020	67323	69020

Notes: The sample includes all candidates who applied in the first wave of the 2012 ASN cycle. Outcomes measure candidates' pre-examination research productivity. *High-impact* articles are published in top-quartile Web of Science journals ranked by AIS in STEM+M or A-list journals in SSH; all other articles are classified as *lower-impact*. *Committee Quality* is committee members' average quality-adjusted research output over the ten years preceding the examination. We instrument realized committee quality with the initially drawn committee's quality net of its expectation. Standard errors are clustered at the exam level.
*** p < .01, ** p < .05, * p < .1

Table A10: First-stage regression for committee quality

	Final Committee Quality	
	(1)	(2)
Initial Committee Quality	0.838***	
	(0.074)	
Expected Committee Quality	0.277	
	(0.257)	
Random Shock to Initial Committee Quality ($\tilde{Z}_e$)		0.831***
		(0.073)
Observations	69020	69020
KP rk Wald F-statistic	129	130
Number of Exams	184	184

Notes: The sample includes all candidates who applied in the first wave of the 2012 ASN cycle. The table reports the first-stage regression for Equation (1), corresponding to column 1 of Table 1. *Committee Quality* is committee members' average quality-adjusted research output over the ten years preceding the examination. *Random Shock to Initial Committee Quality* ($\tilde{Z}_e$) is the initially drawn committee's quality net of its expectation. The specification includes four broad disciplinary group indicators and controls for evaluator–candidate connections. Standard errors are clustered at the exam level.
*** p < .01, ** p < .05, * p < .1

Table A11: Evaluator quality and mentions of bibliometric indicators in reports

	Mentioning Bibliometric Indicator			
	(1)	(2)	(3)	(4)
	Overall	Overall	Negative Vote	Positive Vote
Evaluator Quality	-0.028**	-0.025*	-0.032**	-0.016
	(0.014)	(0.013)	(0.013)	(0.014)
Observations	241669	241669	133450	108219
Candidate Connections	✓	✓	✓	✓
Report Length		✓	✓	✓

Notes: The sample includes all individual evaluator reports for applications submitted in the first wave of the 2012 ASN cycle. The outcome indicates whether the report mentions bibliometric measures or databases, such as the h-index, citations, A-list articles, Scopus, or Web of Science. *Evaluator Quality* measures the evaluator's quality-adjusted research output over the ten years preceding the examination. Columns 1–2 use all reports; columns 3 and 4 restrict the sample to reports accompanying negative and positive votes, respectively. Standard errors are clustered at the exam level.
*** p < .01, ** p < .05, * p < .1

Table A12: Committee quality and qualified candidates' subsequent outcomes: unanimity-restricted sample

	All Candidates					Associate Professor Candidates		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Total Publications	High-Impact Articles	Citations (All Articles)	Citations (Pre-Exam Articles)	Citations (Post-Exam Articles)	Promoted Full Prof.	Future Evaluator	Future Top Evaluator
Panel A. Unanimity Restricted Successful Candidates (without Controls)								
Share of candidates qualified by better than average committee: 0.341								
Share of candidates unanimously qualified by worse than average committee: 0.338								
Above Average Committee	-0.001	0.027	0.046**	0.044*	0.032	0.035**	0.023**	0.006
	(0.024)	(0.025)	(0.023)	(0.025)	(0.020)	(0.014)	(0.009)	(0.005)
Observations	23424	23424	23424	23424	23424	16269	16269	16269
R^2	-0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Baseline	0.26	0.28	0.16	0.20	0.12	0.17	0.11	0.04
Panel B. Unanimity Restricted Successful Candidates (with Controls)								
Share of candidates qualified by better than average committee: 0.341								
Share of candidates unanimously qualified by worse than average committee: 0.338								
Above Average Committee	-0.002	0.011	0.025*	0.011	0.023	0.028**	0.021**	0.006
	(0.017)	(0.019)	(0.015)	(0.012)	(0.015)	(0.012)	(0.008)	(0.004)
Observations	23420	23420	23420	23420	23420	16266	16266	16266
R^2	0.18	0.16	0.37	0.61	0.16	0.02	0.01	0.02
Baseline	0.26	0.28	0.16	0.20	0.12	0.17	0.11	0.04

Notes: The table reports 2SLS estimates of the effect of assignment to a committee of above-expected quality on candidates' subsequent outcomes. The sample includes all candidates qualified by committees with a positive quality shock and only unanimously qualified candidates from committees with a negative quality shock. Columns 1–5 include candidates at both application levels; columns 6–8 include only associate-professor applicants. Panel A controls for discipline fixed effects. Panel B additionally controls for candidate productivity, affiliation, and academic rank; evaluator–candidate connections and research similarity. Standard errors are clustered at the exam level.
*** p < .01, ** p < .05, * p < .1

TABLE A13: CANDIDATES' SUBSEQUENT OUTCOMES BY NUMBER AND DEFINITION OF TOP RESEARCHERS ON THE COMMITTEE

	Top 25%		Top 10%		Top 5%		Top 1%	
	(1) Citations	(2) FP Promotion	(3) Citations	(4) FP Promotion	(5) Citations	(6) FP Promotion	(7) Citations	(8) FP Promotion
Panel A. Qualified candidates								
One Top Researcher	0.027	0.040	0.006	0.043***	0.050***	0.034***	0.093***	0.033*
	(0.035)	(0.045)	(0.020)	(0.012)	(0.018)	(0.012)	(0.030)	(0.018)
	[0.456]	[0.486]	[0.776]	[0.001]	[0.013]	[0.011]	[0.004]	[0.116]
Two Top Researchers	0.020	0.051	0.066***	0.038**	0.054*	0.042***	—	—
	(0.030)	(0.042)	(0.024)	(0.018)	(0.032)	(0.013)		
	[0.509]	[0.305]	[0.016]	[0.053]	[0.170]	[0.014]		
Three or more Top Researchers	0.078**	0.084*	0.126**	0.044**	—	—	—	—
	(0.036)	(0.045)	(0.058)	(0.022)				
	[0.037]	[0.084]	[0.102]	[0.177]				
Panel B. Non-qualified candidates								
One Top Researcher	-0.008	0.004	0.002	-0.001	-0.022*	0.011**	-0.030*	-0.002
	(0.022)	(0.012)	(0.013)	(0.005)	(0.012)	(0.005)	(0.016)	(0.007)
	[0.723]	[0.781]	[0.901]	[0.858]	[0.082]	[0.049]	[0.080]	[0.793]
Two Top Researchers	-0.007	0.004	-0.031*	0.003	0.006	0.011*	—	—
	(0.019)	(0.011)	(0.016)	(0.006)	(0.014)	(0.006)		
	[0.724]	[0.762]	[0.068]	[0.627]	[0.689]	[0.086]		
Three or more Top Researchers	-0.026	0.017	-0.009	0.014	—	—	—	—
	(0.022)	(0.013)	(0.019)	(0.010)				
	[0.252]	[0.211]	[0.594]	[0.219]				
Number of exams with:								
No Top Researchers	13	13	78	78	118	118	162	162
One Top Researcher	54	54	77	77	54	54	22	22
Two Top Researchers	75	75	23	23	12	12	0	0
Three or more Top Researchers	42	42	6	6	0	0	0	0
Baseline mean: qualified	0.15	0.17	0.15	0.17	0.15	0.17	0.15	0.17
Baseline mean: non-qualified	-0.09	0.04	-0.09	0.04	-0.09	0.04	-0.09	0.04

Notes: The sample includes associate-professor applicants in the first wave of the 2012 ASN cycle: 17,541 qualified and 29,885 rejected candidates. Estimates follow a nonlinear specification of Equation (2). Outcomes are cumulative post-exam citations, standardized within examinations, and an indicator for promotion to full professor within ten years of evaluation. Each pair of columns defines *top researchers* using a different threshold: the top 25%, 10%, 5%, or 1% of quality-adjusted research output among Italian full professors within their discipline over the preceding ten years. The explanatory variables indicate whether the final committee includes exactly one, exactly two, or at least three top researchers; the omitted category is none. All specifications control for the probabilities of drawing each number of top researchers; candidates' pre-evaluation publication counts, publication impact, and citations; broad disciplinary group, candidate institution, and academic rank indicators; and evaluator–candidate connections. Standard errors clustered at the examination level are in parentheses; wild-cluster bootstrap p-values are in square brackets. Significance stars are based on clustered standard errors.

*** p < .01, ** p < .05, * p < .1

Table A14: Randomization check: evaluator characteristics

	Publications		Articles		High-impact Articles		Citations		Academic age	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Drawn researcher	-0.009		-0.012		0.001		0.013		0.207	
	(0.024)		(0.024)		(0.024)		(0.025)		(0.207)	
Drawn researcher (Scaled)		-0.000		-0.004		-0.000		0.032		0.270
		(0.062)		(0.062)		(0.068)		(0.057)		(0.629)
Observations	15712	15712	15712	15712	15712	15712	15712	15712	15712	15712

Notes: The sample includes potential ASN evaluators in 2012–2021. Outcomes measure researchers' pre-examination characteristics. Research output is accumulated over the ten years preceding the examination and standardized within fields. Academic age is measured in years since the researcher's first publication. *High-impact* articles are published in top-quartile Web of Science journals ranked by AIS in STEM+M or A-list journals in SSH. We instrument committee service with the initial draw indicator net of the probability of being drawn. Standard errors are clustered at the exam level.
*** p < .01, ** p < .05, * p < .1

Table A15: First-stage regression for committee service

	Final Draw Status
	(1)
Random Shock to Initial Draw Status ($\tilde{Z}_{j,e}$)	0.888***
	(0.007)
Number of Publications	0.001
	(0.002)
High-Impact Articles	-0.003
	(0.002)
Observations	15712
KP rk Wald F-stat	16497
Number of Exams	685

Notes: The sample includes potential ASN evaluators in 2012–2021. The table reports the first-stage regression for Equation (4), corresponding to the model in column 1 of Figure 1. *Random Shock to Initial Draw Status* ($\tilde{Z}_{j,e}$) is the initial draw indicator net of the researcher's probability of being drawn. The specification controls for examination fixed effects, past research output, and a quadratic in academic age. Standard errors are clustered at the exam level.
*** p < .01, ** p < .05, * p < .1

Table A16: Committee service and research output: binary and team-size-scaled effects

	Accumulated Total Works 3 years post-exam		
	(1)	(2)	(3)
Drawn Researcher	-0.051***		0.007
	(0.017)		(0.026)
Drawn Researcher (Scaled)		-0.187***	-0.202***
		(0.045)	(0.070)
Observations	15712	15712	15712
Adjusted R^2	0.49	0.49	0.49

Notes: The sample includes potential ASN evaluators in 2012–2021. The outcome is cumulative research output over the first four years following assignment. The specification jointly includes *Drawn Researcher*, a committee-service indicator, and *Drawn Researcher (Scaled)*, the same indicator divided by the researcher's average number of authors per paper. These variables are instrumented with the initial draw indicator net of the probability of being drawn and its counterpart divided by team size, respectively. Controls include examination fixed effects, inverse team size, past research output, and a quadratic in academic age. Standard errors are clustered at the exam level.
*** p < .01, ** p < .05, * p < .1

Table A17: Effects of committee service on research output and subsequent volunteering by peer quality

	Panel A. Accumulated Total Works 3 years post-exam				Panel B. Volunteer Again			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Drawn Researcher (Scaled)	-0.187***	-0.172***	-0.175***	-0.175***	-0.077**	-0.086**	-0.067	-0.067
	(0.045)	(0.057)	(0.057)	(0.057)	(0.032)	(0.043)	(0.045)	(0.045)
Evaluator Quality	0.058***	0.103***	0.102***	0.103***	0.004	-0.001	-0.004	-0.004
	(0.013)	(0.015)	(0.015)	(0.015)	(0.008)	(0.011)	(0.011)	(0.011)
Drawn Researcher (Scaled) × Evaluator Quality		-0.135**	-0.132**	-0.129**		-0.074**	-0.073**	-0.073**
		(0.057)	(0.055)	(0.054)		(0.035)	(0.035)	(0.035)
Drawn Researcher (Scaled) × Peer Quality				0.035				0.012
				(0.096)				(0.076)
Observations	15712	15712	15712	15712	8696	8696	8696	8696
Exam Fixed Effects	✓	✓			✓	✓		

Notes: The sample includes potential ASN evaluators in 2012 and 2016. Panel A measures cumulative research output over the three years following the examination year, standardized within examinations. Panel B measures volunteering in 2018 for the 2012 cohort and in 2023 for the 2016 cohort. It excludes researchers drawn in intervening cycles (2016 and 2018–2021, respectively), since such service makes them ineligible to volunteer in the subsequent cycle. *Evaluator Quality* (T_j) measures quality-adjusted research output over the ten years preceding the examination. *Peer Quality* ($T_{e,-j}$) is the corresponding average among fellow evaluators. All specifications control for four broad disciplinary groups, the probability of being drawn, and its interactions with expected peer quality. Standard errors are clustered at the exam level.
*** p < .01, ** p < .05, * p < .1

B Data sources and construction

We construct the main database by combining four sources: (i) CINECA records on professors employed at Italian public universities, (ii) official ASN records on eligible evaluators, and committee draws, (iii) publication records from IRIS and OpenAlex, and (iv) candidate-level data collected by Bagues et al. (2017). This appendix summarizes the construction of researcher identifiers, the linkage of publication records, and the simulation of the ASN draw procedure. We also describe information we identified on teaching reductions to evaluators.

B.1 Data on ASN records and publications

Researcher identifiers. CINECA provides the backbone of the researcher database because it covers the universe of professors employed at Italian public universities. The data include each professor's name, gender, university, department, research area, and academic rank. We assign each professor a unique `cineca_id` that tracks individuals over time. In almost all cases, names uniquely identify professors within a year. For the small number of homonyms, we disambiguate using research area and, where necessary, university affiliation; the remaining cases are checked manually.

We then link ASN evaluator and candidate records to CINECA. Evaluators are matched primarily by name within macro field, after standardizing names and correcting common typographical and OCR inconsistencies in the Ministry files. Because eligible evaluators are drawn from the population of Italian full professors within each field, the risk of false matches is limited. This procedure links virtually all eligible evaluators to CINECA; only eight researchers in the evaluator pool remain unmatched.

Candidates are matched to CINECA using a more conservative procedure, because some candidates apply from outside the Italian university system and are therefore not observed in CINECA. We standardize names as for evaluators, but exclude riskier match types and require a match on both given and family names. We also exclude full professors from the set of potential CINECA matches, since ASN candidates apply for associate or full professorship qualification. This procedure links 33,446 of 46,329 candidates, or approximately 72 percent, to unique CINECA identifiers. Unmatched candidates are mainly researchers outside the Italian public university system.

Finally, we construct a master researcher identifier, `id`, that covers individuals appearing in CINECA, the ASN evaluator files, or the ASN candidate files. When a researcher is observed in

multiple sources, we prioritize the CINECA identifier, then the evaluator identifier, and then the candidate identifier. We harmonize name, gender, affiliation, and field information within each `id`, retaining one standardized researcher record for merging with publication data.

Publication records. We link researchers to publication data from IRIS and OpenAlex. IRIS is maintained by Italian universities and contains self-reported publication records of researchers affiliated with Italian institutions. We use two scrapes, from 2020 and 2024, and remove duplicate author-publication records within and across scrapes.[37] IRIS author profiles are matched to researchers using standardized names and university affiliations. Exact and unique matches are accepted directly; non-unique names are resolved using affiliation histories. Ambiguous matches are discarded.

We complement IRIS with OpenAlex, which provides broader publication coverage and richer bibliographic metadata. The OpenAlex linkage begins with a broad set of name-based candidate matches, allowing for common differences in name order, initials, middle names, particles, and compound surnames. We then retain only links supported by additional evidence. Candidate pairs are confirmed when the name match is unique and is supported by at least one independent signal, such as affiliation history, journal overlap, country, or discipline. Confirmed OpenAlex profiles are removed from the candidate set for all other researchers before weaker rules are applied, preventing the same profile from being assigned to multiple individuals. Ambiguous or implausible links are discarded.

We combine IRIS and OpenAlex into a unified publication database. We harmonize publication types and retain substantive outputs, including journal articles, books, book chapters, and conference proceedings. Duplicate publications across sources are identified using cleaned titles, publication years, DOIs, ISSNs, journal names, and overlapping author information. When the same publication appears in both databases, we retain the OpenAlex record because it typically contains richer metadata and author identifiers. The resulting database contains one record per publication and forms the basis for the productivity and impact measures used in the analysis.

IRIS covers 80% of professors in our administrative data and 88% of potential evaluators; OpenAlex covers 89% and 90%, respectively. Combining the two sources increases coverage to 96% of professors and approximately 92% of ASN candidates. If a candidate or evaluator cannot be

37. We collected these records from universities' IRIS portals. The most recent collection was in December 2024. The portals share a similar interface; see, for example, Sapienza University of Rome at https://iris.uniroma1.it/, last accessed September 2025.

matched to publication records, we assign zero to their publication-based measures.

Validating OpenAlex and IRIS database We begin by reporting summary statistics on candidate research profiles. We find about 92% of candidates in OpenAlex and IRIS. In *OpenAlex+IRIS*, the average candidate has 29 matched publications, of which 28 are journal articles; 43% of these journal articles appear in high-impact journals (Appendix Tables B1). This appears to reflect the fact that OpenAlex and IRIS provide less comprehensive coverage of non-journal research output. Specifically, we fail to capture the number of books and chapters written by candidates. At the same time, the mean number of articles and high-impact articles are remarkably similar across the two databases, suggesting that the combined database does well at capturing journal-based output.

The correlation between the number of journal articles in candidates' curricula vitae and in *OpenAlex+IRIS* is 78% (Table B2), suggesting that the combined external database captures journal-based output reliably. Correlations are lower for books and chapters, which are less consistently indexed in external publication databases.

Table B1: Comparison of candidate research output across CVs and OpenAlex–IRIS data

	Mean	SD
Total Publications (CV)	47.16	42.38
Total Publications (OA+IRIS)	28.02	35.14
Articles (CV)	25.97	30.17
Articles (OA+IRIS)	25.19	31.75
High-Impact Articles (CV)	10.68	16.45
High-Impact Articles (OA+IRIS)	10.75	17.09
Books (CV)	1.89	3.36
Books (OA+IRIS)	0.17	1.06
Chapters (CV)	5.87	9.19
Chapters (OA+IRIS)	0.46	1.95
Number of Candidates	46329	
Number of Candidates Matched in OpenAlex and IRIS	42472	

Notes: The sample includes candidates in the first wave of the 2012 ASN cycle matched to the combined OpenAlex–IRIS database. Research output is accumulated over the ten years preceding the examination. *High-impact* articles are published in top-quartile Web of Science journals ranked by AIS in STEM+M or A-list journals in SSH.

Table B2: Correlations between candidate research measures in CVs and OpenAlex–IRIS data.

	CV Measures				
	Total Publications	Articles	High-Impact Articles	Books	Chapters
Total Publications	.67	.	.	.	.
Articles	.	.8	.	.	.
High-Impact Articles	.	.	.81	.	.
Books	.	.	.	.05	.
Chapters	.	.	.	.	.06

Notes: The sample includes candidates in the first wave of the 2012 ASN cycle matched to the combined OpenAlex–IRIS database. Research output is accumulated over the ten years preceding the examination. *High-impact* articles are published in top-quartile Web of Science journals ranked by AIS in STEM+M or A-list journals in SSH.

B.2 Simulation of ASN draw probabilities.

We simulate the ASN draw procedure to recover each eligible evaluator's *ex ante* probability of being selected and the *ex ante* probability that any pair of evaluators serves on the same committee. These probabilities are used to control for expected committee composition and expected peer quality.

For the 2012 ASN, the draw procedure imposed two main constraints: representation across research areas and at most one evaluator per university. SSDs were ordered by the number of eligible volunteers, with smaller SSDs considered first. One evaluator was drawn from each large SSD, defined as an SSD with more than 30 researchers in CINECA, subject to the university constraint. Remaining committee seats were then filled from the full pool of eligible evaluators, again respecting the constraint that no two selected evaluators came from the same university.

For the 2016–2023 ASN rounds, the same principles applied, but the rules for SSD representation changed. The threshold for guaranteed representation was reduced to 10 eligible full professors, and the number of guaranteed seats became proportional to SSD size. For each discipline, we compute the expected number of evaluators from SSD s as

$$E_s = \frac{N_s}{\sum_{s' \in S} N_{s'}} \times N,$$

where N_s is the number of eligible professors in SSD s, S is the set of SSDs in the discipline, and N is the number of evaluators drawn for the exam. The algorithm first fills the guaranteed SSD-specific seats, respecting the university constraint, and then fills any remaining seats from the full pool of eligible evaluators.

For each ASN round, we replicate the relevant draw algorithm one million times. The simu-

lated draw probabilities are used in the empirical analysis to separate realized random variation in committee composition from predictable differences generated by the structure of the eligible pool and the draw rules.

B.3 Documentary evidence on teaching relief

We searched publicly available university regulations, governing-body minutes, and teaching plans for references to teaching relief associated with ASN committee service. The records provide examples of institutional responses to evaluation duties, rather than a representative measure of relief across universities.

The 2012 ASN call permits partial teaching exemption upon request, without additional public expenditure.[38] At Bari, departmental minutes dated January 30, 2013 record approval of a 100-hour reduction from a 120-hour teaching assignment for an ASN commissioner in 2012–13.[39] At Macerata, teaching plans considered by the Academic Senate on April 16, 2019 list three ASN commissioners with 75 teaching hours against a normal obligation of 120, noting that each had requested a reduction.[40] Tor Vergata's Department of Economics and Finance established a 40-hour reduction for ASN commissioners in December 2025, applicable to requests for 2025–26 and subsequent requests.[41]

Relief was not always requested or granted. Minutes of the University of Padova's FISPPA departmental council dated February 25, 2013 note that other eligible evaluators did not intend to request teaching relief. The department sought to accommodate the request under consideration through an unpaid teaching assignment, consistent with the requirement that relief entail no additional public expenditure.[42] At the University of Palermo, the Academic Senate rejected a request for teaching relief associated with continued service as an ASN committee chair on December 17, 2013.[43]

These records do not establish how frequently relief was granted or how teaching reductions translated into research time preserved. Accordingly, the workload and research-time adjustments used in the main text are illustrative assumptions rather than estimates derived from these docu-

38. Decreto Direttoriale n. 181, June 27, 2012, Article 8(4).

39. University of Bari, FOR.PSI.COM departmental council minutes, January 30, 2013, p. 11.

40. University of Macerata, Academic Senate documents, April 16, 2019, Annex 9, Table B.

41. University of Rome Tor Vergata, Department of Economics and Finance council minutes, December 16, 2025, pp. 4–5, item 3.1.0.

42. University of Padova, FISPPA departmental council minutes, February 25, 2013.

43. University of Palermo, Academic Senate decisions, December 17, 2013, item 6.

ments.

C Robustness Checks: Alternative Measures of Research Profiles

In this section, we validate the publication database compiled from OpenAlex and IRIS by comparing it to candidates' self-reported curricula vitae, we describe other dimensions of candidate research profiles, and perform a series of robustness checks to make sure our results are not being driven by decisions in how we construct our measures.

C.1 Candidate research profiles

In the main analysis, we characterize candidates' research profiles using publication quantity and journal-based impact. These measures capture distinct dimensions of research output and are only moderately correlated. To assess whether our findings depend on this choice, we consider alternative measures of impact and additional characteristics related to collaboration, authorship, career stage, and thematic content. Appendix Table C1 summarizes these measures. All measures are standardized across candidates within the same exam.

Appendix Table C2 reports the correlations among these dimensions, while Appendix Table C3 examines their association with qualification outcomes. When considered individually, publication quantity, journal-based impact, citations, and novelty are positively associated with qualification, whereas academic age, topic dispersion are negatively associated with qualification. The share of first-, last-, or sole-authored publications has no unconditional association with qualification. One possible explanation is that researchers in large, high-impact collaborations are less likely to occupy these authorship positions.

Among the indicators considered individually, publication quantity and journal-based impact measures have the greatest explanatory power, as measured by the adjusted R^2, supporting their use as the core dimensions in the main analysis. When the measures enter jointly – including novelty for STEM+M fields in column 10 and excluding it for the full sample in column 11 – journal-based impact remains a strong predictor of qualification. Its coefficient increases, while the coefficient on citations attenuates substantially.

C.2 Committee quality and returns to different dimensions of candidate research profiles

Following the logic of the analysis in Section 4, we examine whether committee quality affects the estimated returns to additional dimensions of candidates' research profiles, beyond the quantity and impact measures analyzed in the main text. Consistent with the main analysis, committee quality appears to significantly amplify the returns to publication impact (Appendix Table C4). In contrast, the returns to publication quantity are no longer significantly affected by committee quality, likely reflecting a correlated negative effect of committee quality on seniority, which is now controlled for in the regression. The returns to other dimensions of candidates' research profiles, conditional on these effects, do not appear to be significantly influenced by committee quality.

C.3 Robustness to the choice of measures of candidate publication quantity and impact

We begin by confirming that our results (reproduced in Column 1 of Appendix Table C5) are robust to using alternative data sources. The same analysis based on OpenAlex and IRIS publication data explains less variation in the outcome variable, as reflected in a lower Adjusted R^2, and we observe smaller estimated returns to publication impact (Column 2 of Appendix Table C5). This is likely because the alternative data no longer exploit information directly provided to evaluators. The key coefficient of interest – the interaction between *Candidate's Publication Impact* and *Committee Quality* – is indistinguishable from the estimate in the main text, with an identical (albeit more noisy) point estimate.

Column 3 reports results using an alternative measure of *Quantity*, scaled by the candidate's average number of coauthors. This adjustment accounts for the possibility that evaluators place less weight on publications produced by larger teams. The R^2 is lower in this specification, suggesting that this variable is less informative about evaluators' decision-making process. All our results remain quantitatively unchanged.

Next, we test the robustness of our findings to alternative measures of publication impact. First, we assess whether defining impact as the *share* of high-impact articles drives our results. When candidates' publication impact is instead measured as the *number* of high-impact articles, the coefficients on the key variables of interest are statistically indistinguishable from those obtained using the share-based measure (Column 4 of Appendix Table C5). Notably, the marginal effect of publication quantity declines, while the marginal effect of publication impact increases substan-

tially. This likely reflects that the count-based impact measure partially captures non-linearities in publication quantity. Since the results are not affected by this modeling choice – and because the share measure is easier to interpret in combination with a quantity measure – we use the share of high-impact publications in the main analysis.

C.4 Robustness to alternative measures of evaluator quality

Our baseline measure of evaluator quality is the number of high-impact articles published before the exam. We assess the sensitivity of our findings to alternative measures, summarized in Appendix Table C6.

Appendix Table C7 shows that the qualitative patterns are consistent across measures, although statistical significance varies. The coefficient on *Committee Quality* is negative throughout and statistically significant at the 10% level or better for the baseline, top-researcher, AIS, and Web of Science article measures. The interaction with *Candidate's Number of Publications* is negative in every specification and statistically significant at the 5% level except for the AIS and Web of Science article measures. Conversely, the interaction with *Candidate's Publication Impact* is consistently positive, but statistically significant at the 10% level for total-publication, article, and AIS measures. Overall, these results support the interpretation that better-published committees are stricter and place greater weight on publication impact relative to quantity, although some alternative measures provide less precise evidence.

Table C1: Definitions of candidate research measures

Measure	Definition
Panel A. Measures used in the main analysis	
Publication quantity	Total number of publications prior to the exam, including journal articles, books, book chapters, and conference proceedings.
Journal-based impact	Share of journal articles published in top-quartile Web of Science journals ranked by AIS in STEM+M or ANVUR-designated A-list journals in SSH.
Panel B. Alternative measures of impact	
Article Influence Score (AIS)	Journal-based impact measured using the Article Influence Score of the outlets in which the candidate publishes.
Citations per publication	Average number of citations per publication.
Panel C. Additional profile characteristics	
Coauthors per publication	Average number of coauthors per publication.
First-, last-, or sole-authored share	Share of publications on which the candidate is listed as first, last, or sole author.
Seniority	Year of the candidate's first recorded publication.
Topic dispersion	Number of distinct OpenAlex topics represented in the candidate's publications, divided by the total number of publications.
Novelty	Index measuring new combinations of words or phrases in the literature, following Arts et al. (2025). Available for STEM+M fields only.

Notes: Definitions refer to measures before standardization. All measures are standardized within examinations.

Reference: Arts, S., N. Melluso, and R. Veugelers (2025). "Beyond Citations: Measuring Novel Scientific Ideas and their Impact in Publication Text." *The Review of Economics and Statistics*, https://doi.org/10.1162/rest_a_01561.

Table C2: Correlation between dimensions of candidate research profiles

	Quantity	Journal Impact	AIS	Citations	Collaborators per Paper	Seniority	Topic Dispersion	FirstLastSingle Author per Paper	Novelty
Quantity	1	-.09	.32	.13	.06	-.1	-.2	-.05	.02
Journal Impact	-.09	1	.29	.13	.14	.1	-.08	-.1	.15
AIS	.32	.29	1	.3	.27	.02	-.15	-.15	.1
Citations	.13	.13	.3	1	.1	-.09	-.28	-.06	.11
Collaborators per Paper	.06	.14	.27	.1	1	.09	-.05	-.5	.14
Seniority	-.1	.1	.02	-.09	.09	1	.05	-.06	.02
Topic Dispersion	-.2	-.08	-.15	-.28	-.05	.05	1	.03	-.05
FirstLastSingle Author per Paper	-.05	-.1	-.15	-.06	-.5	-.06	.03	1	-.1
Novelty	.02	.15	.1	.11	.14	.02	-.05	-.1	1

Notes: The sample includes all candidates who applied in the first wave of the 2012 ASN cycle. Correlations involving *Novelty* are restricted to STEM+M fields, where the measure is more reliable. All measures are standardized within examinations and defined in Table C1.

Table C3: Candidate research profiles and qualification

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)
Quantity	0.113***									0.077***	0.096***
	(0.005)									(0.008)	(0.005)
Impact: Journal Impact		0.082***								0.052***	0.080***
		(0.005)								(0.005)	(0.004)
Impact: Article Influence Score			0.095***							0.064***	0.037***
			(0.008)							(0.012)	(0.007)
Impact: Citations				0.062***						0.033***	0.016***
				(0.005)						(0.005)	(0.003)
Collaborators per Paper					0.001					-0.032***	-0.024***
					(0.004)					(0.005)	(0.004)
Seniority						-0.004***				-0.000	-0.003***
						(0.001)				(0.001)	(0.001)
Topic Dispersion							-0.091***			-0.068***	-0.052***
							(0.003)			(0.006)	(0.003)
First+Last+Single Author per Paper								0.001		0.015***	0.008**
								(0.004)		(0.005)	(0.003)
Novelty									0.006	-0.010***	
									(0.004)	(0.004)	
Observations	69020	69020	69020	69020	69020	69020	69020	69020	34596	34596	69020
Adjusted R^2	0.055	0.028	0.038	0.017	-0.000	0.004	0.029	-0.000	0.000	0.132	0.113

Notes: The sample includes all candidates who applied in the first wave of the 2012 ASN cycle; columns 9–10 are restricted to STEM+M fields. The outcome indicates whether the candidate qualified. All research measures are standardized within examinations and defined in Table C1. Standard errors are clustered at the exam level.
*** p < .01, ** p < .05, * p < .1

Table C4: Committee quality and the association between candidate research profiles and qualification

	STEM (1)	Overall (2)
Quantity	0.075***	0.096***
	(0.008)	(0.005)
Impact: Journal Impact	0.052***	0.080***
	(0.005)	(0.004)
Impact: Article Influence Score	0.065***	0.036***
	(0.012)	(0.007)
Impact: Citations	0.033***	0.016***
	(0.005)	(0.003)
Collaborators per Paper	-0.031***	-0.023***
	(0.005)	(0.004)
Seniority	-0.000	-0.003***
	(0.001)	(0.001)
Topic Dispersion	-0.067***	-0.052***
	(0.006)	(0.003)
First+Last+Single Author per Paper	0.016***	0.008**
	(0.004)	(0.003)
Novelty	-0.011***	
	(0.003)	
Quantity × Committee Quality	-0.034	-0.027*
	(0.020)	(0.014)
Impact: Journal Impact × Committee Quality	0.008	0.027***
	(0.010)	(0.009)
Impact: Article Influence Score × Committee Quality	0.020	0.012
	(0.024)	(0.017)
Impact: Citations × Committee Quality	0.009	-0.005
	(0.011)	(0.008)
Collaborators per Paper × Committee Quality	0.013	0.008
	(0.013)	(0.010)
Seniority × Committee Quality	0.002	0.003
	(0.002)	(0.002)
Topic Dispersion × Committee Quality	0.015	-0.001
	(0.015)	(0.008)
FirstLastSingle Author per Paper × Committee Quality	0.013	0.005
	(0.010)	(0.009)
Novelty × Committee Quality	-0.016**	
	(0.008)	
Observations	34596	69020
Number of Exams	109	184
Adjusted R^2	0.14	0.12
Candidate Connections	✓	✓

Notes: The sample includes candidates who applied in the first wave of the 2012 ASN cycle. Column 1 is restricted to STEM+M fields; column 2 includes all fields. The outcome indicates whether the candidate qualified. All candidate research measures are standardized within examinations and defined in Table C1. All specifications include broad disciplinary group indicators and controls for evaluator–candidate connections through coauthorship, shared affiliation, and shared subfield, as well as research similarity with the committee. Standard errors are clustered at the exam level.
*** p < .01, ** p < .05, * p < .1

Table C5: Committee quality and qualification: alternative sources and definitions of candidate research measures

	Main-Text	OpenAlex IRIS	OpenAlex IRIS - Scaled	Count High-Impact
	(1)	(2)	(3)	(4)
Candidate Quantity Measure	0.121***	0.118***	0.113***	0.064***
	(0.004)	(0.004)	(0.004)	(0.006)
Candidate Impact Measure	0.093***	0.061***	0.066***	0.111***
	(0.005)	(0.004)	(0.004)	(0.008)
Committee Quality	-0.054*	-0.054*	-0.054*	-0.054*
	(0.029)	(0.029)	(0.029)	(0.029)
Candidate Quantity Measure x Committee Quality	-0.026**	-0.015	-0.013	-0.045***
	(0.012)	(0.010)	(0.008)	(0.014)
Candidate Impact Measure x Committee Quality	0.035***	0.034***	0.033***	0.042**
	(0.012)	(0.010)	(0.010)	(0.017)
Observations	69020	69020	69020	69020
Adjusted R^2	0.094	0.071	0.066	0.101

Notes: The sample includes all candidates who applied in the first wave of the 2012 ASN cycle. The outcome indicates whether the candidate qualified. *Committee Quality* is evaluators' average quality-adjusted research output over the ten years preceding the examination. Candidate research measures are standardized within examinations; evaluator quality is standardized within eligibility pools. We instrument realized committee quality with the initially drawn committee's quality net of its expectation under the assignment mechanism. Columns 1 and 4 use candidate publication data from CVs; columns 2–3 use OpenAlex–IRIS data. Column 1 reports the baseline specification. Column 3 scales publication quantity by average team size. Column 4 measures publication impact by the number rather than the share of high-impact articles. All specifications include broad disciplinary group indicators and controls for evaluator–candidate connections and research similarity. Standard errors are clustered at the exam level.
*** p < .01, ** p < .05, * p < .1

Table C6: Definitions of evaluator research measures

Measure	Definition
Panel A. Measure used in the main analysis	
High-impact articles	Number of articles published in top-quartile Web of Science journals ranked by AIS in STEM+M or ministry-designated A-list journals in SSH.
Panel B. Alternative measures	
Total publications	Total number of publications, including journal articles, books, book chapters, and conference proceedings.
Articles	Total number of journal articles.
AIS	Sum of journal articles, each weighted by its journal's Article Influence Score.
Articles WOS	Number of articles published in journals indexed in Web of Science.
Citations	Total citations received by the evaluator's publications.

Notes: All measures cover the ten years preceding the examination.

TABLE C7: COMMITTEE QUALITY AND QUALIFICATION: ALTERNATIVE MEASURES OF EVALUATOR RESEARCH ACHIEVEMENT

	Main-Text	Top Researcher	Total Publications	Articles	AIS	Articles WOS	Citations
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Candidate's Number of Publications	0.121***	0.148***	0.121***	0.121***	0.120***	0.120***	0.121***
	(0.004)	(0.014)	(0.004)	(0.004)	(0.004)	(0.004)	(0.004)
Candidate's Impact of Publications	0.093***	0.069***	0.093***	0.093***	0.094***	0.093***	0.093***
	(0.005)	(0.015)	(0.005)	(0.005)	(0.005)	(0.005)	(0.005)
Committee Quality	-0.054*	-0.173***	-0.036	-0.043	-0.055**	-0.050*	-0.014
	(0.029)	(0.062)	(0.027)	(0.028)	(0.027)	(0.029)	(0.032)
Candidate's Number of Publications x Committee Quality	-0.026**	-0.060**	-0.024**	-0.025**	-0.017	-0.017	-0.018**
	(0.012)	(0.029)	(0.012)	(0.013)	(0.012)	(0.012)	(0.009)
Candidate's Impact of Publications x Committee Quality	0.035***	0.053	0.027*	0.026*	0.024*	0.020	0.010
	(0.012)	(0.032)	(0.014)	(0.015)	(0.012)	(0.013)	(0.013)
Observations	69020	69020	69020	69020	69020	69020	69020
Adjusted R^2	0.094	0.091	0.093	0.094	0.094	0.094	0.092

Notes: The sample includes candidates who applied in the first wave of the 2012 ASN cycle. The outcome indicates whether the candidate qualified. Alternative evaluator research measures are defined in Table C6. Candidate research measures are standardized within examinations; evaluator quality is standardized within eligibility pools before computing committee quality (T_e). *Top researchers* are in the top quartile of quality-adjusted research output among Italian full professors within their field over the ten years preceding the examination. We instrument final committee composition with the initial draw, before resignations. All specifications include broad disciplinary group indicators and controls for evaluator–candidate connections through coauthorship, shared affiliation, and shared subfield, as well as research similarity with the committee. Standard errors are clustered at the exam level.

*** p < .01, ** p < .05, * p < .1

D The marginal treatment effect of qualification

This appendix examines whether heterogeneous effects of qualification could explain the subsequent outcome differences between candidates assessed by better- and worse-published committees. We exploit variation in candidate–committee connections and research proximity generated by random committee assignment.

We first estimate the effect of qualification by 2SLS, instrumenting qualification with unexpectedly strong connections to the committee. We construct this instrument by subtracting the connections expected given the eligible evaluator pool and the rules governing the draw (Borusyak and Hull, 2023). The estimates capture effects for candidates whose qualification status responds to this variation. Table D1 reports the first stage, and Table D2 reports the outcome estimates. Among associate professor applicants, qualification raises promotion within five years by 43 percentage points but reduces subsequent research productivity. Among full professor applicants, qualification also facilitates promotion, while increasing publication in high-impact journals.

We next estimate marginal treatment effects following Heckman and Vytlacil (2007) and Carneiro et al. (2011). Let D_i denote qualification, X_i observed candidate characteristics and examination controls, and Z_i the excluded connection and research-proximity instruments. We assume the scalar latent-index representation

$$D_i = \mathbf{1}\{P(Z_i, X_i) \geq U_i\}, \tag{D1}$$

where $P(Z_i, X_i) = \Pr(D_i = 1 \mid Z_i, X_i)$ is the qualification propensity score and U_i is unobserved resistance to qualification, normalized to be uniform conditional on X_i. This representation imposes monotonicity: instrument changes that raise the propensity score weakly increase qualification for every candidate, preserving candidates' ordering by unobserved resistance.

Identification requires

$$(Y_{1i}, Y_{0i}, U_i) \perp Z_i \mid X_i, \tag{D2}$$

where Y_{1i} and Y_{0i} denote potential outcomes with and without qualification. Random assignment provides the source of exogenous variation, conditional on the assignment mechanism. Exclusion additionally requires that connections and research proximity affect subsequent outcomes only through qualification. The institutional separation between ASN qualification and university appointments supports this restriction: ASN committees have no formal role in subsequent university-level evaluations, and qualification confers the same eligibility regardless of how it was obtained.

The marginal treatment effect is

$$\begin{aligned} \mathrm{MTE}(x,u) &= \mathbb{E}\big[Y_{1i} - Y_{0i} \mid X_i = x, U_i = u\big] \\ &= \frac{\partial\, \mathbb{E}[Y_i \mid X_i = x, P(Z_i, X_i) = p]}{\partial p}\bigg|_{p=u}. \end{aligned} \tag{D3}$$

It measures the effect of qualification for candidates at different points of the qualification margin. Identification requires instrument-induced propensity-score variation conditional on observables; the derivative is identified where this conditional support permits local variation.

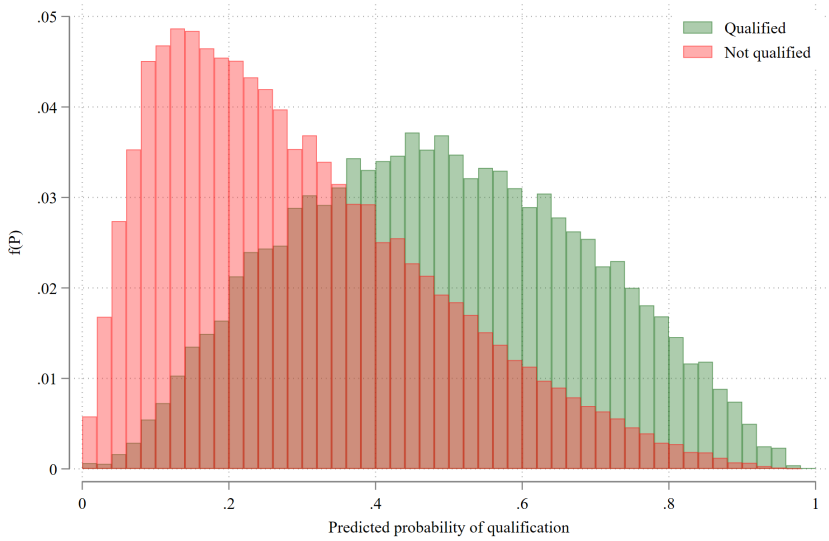
We estimate the propensity score using a logit model with candidate characteristics, examination fixed effects, and connection variables. The estimates confirm that connections significantly affect the probability of qualification and generate substantial variation in the propensity score (Table D1). Figure D1 shows substantial overlap between qualified and rejected candidates over $p \in [0.1, 0.8]$. The outcome specification includes second-order polynomials in the propensity score and candidate characteristics, together with their interactions. We evaluate the implied MTE at 30 propensity-score values, holding observables at their means, and bootstrap both stages.

For associate professor applicants, the estimated MTE of qualification on subsequent citations rises from negative values at low resistance to positive values at higher resistance, but is imprecisely estimated and nowhere significantly positive (Figure D2). The positive slope may reflect mean reversion among candidates with unusually high pre-exam output or a shift toward administrative and service responsibilities following qualification.

Qualification accelerates promotion to associate professor among candidates with low resistance (Figure D3, Panel A), while the estimates suggest limited heterogeneity in its effect on subsequent promotion to full professor (Panel B).

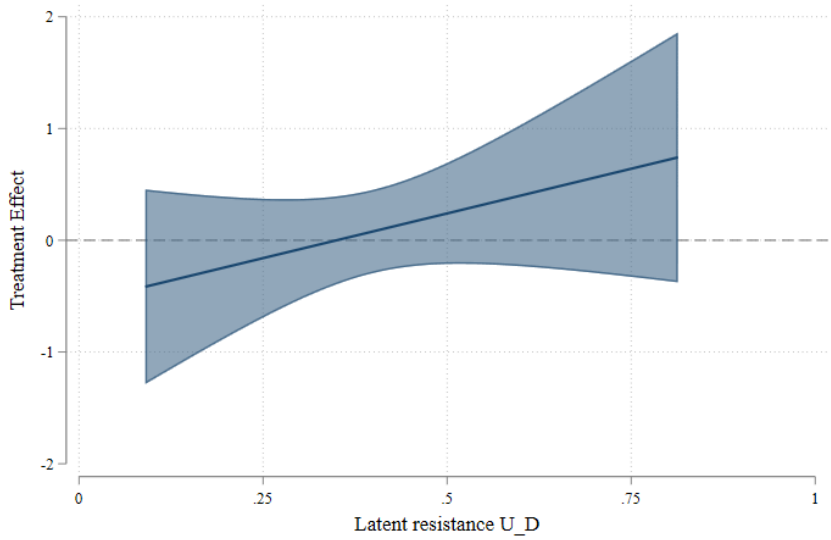
The citation estimates provide little support for heterogeneous returns to qualification as an explanation for our main findings. In particular, the estimated return is negative at low resistance to qualification, suggesting that qualification does not necessarily improve subsequent research outcomes for candidates near this margin. Denying qualification to such candidates would therefore tend to improve, rather than worsen, their citation outcomes. This runs against an explanation in which the weaker subsequent performance of rejected candidates reflects the loss of benefits from qualification. Although the estimates are imprecise, their pattern supports the interpretation that better-published committees distinguish candidates with stronger and weaker research prospects.

Figure D1: Support of estimated success probability by qualification status



Notes: The figure displays the support of the estimated propensity score for qualified and non-qualified candidates. The propensity score is obtained from the logit model reported in Table D1. Each bar represents the fraction of observations within a given bin of the propensity score.

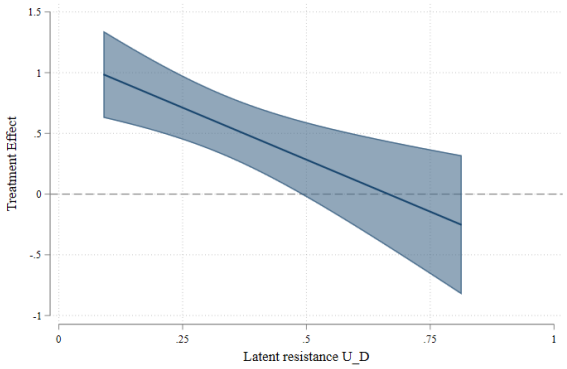
Figure D2: Marginal Treatment Effect of Qualification on Future Citations



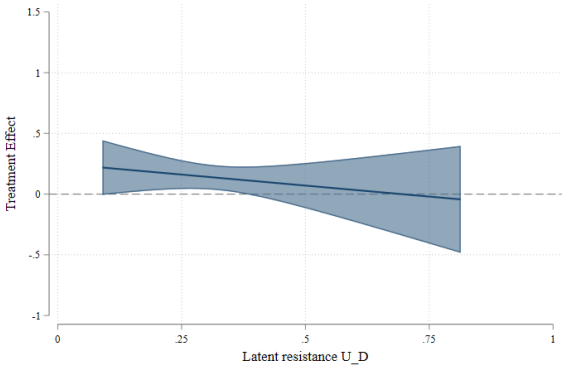
Notes: The figure reports the marginal treatment effect of qualification to associate professor on the cumulative number of future citations. The measure is normalized within examinations. To estimate the function plotted here, we first estimate a partially linear specification of the promotion outcome on flexible functions of the predicted probability of qualification, $g(P)$, a second order polynomial in covariates $f(X)$, and interactions between P and $f(X)$. The MTE curve is obtained by evaluating the derivative of the estimated conditional expectation function with respect to P at the average value of X. The solid line reports the estimated MTE; the shaded area reports the 90 percent bootstrap confidence interval (250 replications).

Figure D3: Marginal Treatment Effect of associate professor Qualification on Promotion

Panel A.
Promoted to Associate Prof. within 5 years



Panel B.
Promoted to Full Prof. within 10 years



Notes: The figure reports the marginal treatment effect of qualification to associate professor on the probability of subsequent promotions. To estimate the function plotted here, we first estimate a partially linear specification of the promotion outcome on flexible functions of the predicted probability of qualification, $g(P)$, a second order polynomial in covariates $f(X)$, and interactions between P and $f(X)$. The MTE curve is obtained by evaluating the derivative of the estimated conditional expectation function with respect to P at the average value of X. The solid line reports the estimated MTE; the shaded area reports the 90 percent bootstrap confidence interval (250 replications).

Table D1: Connections and qualification: first-stage logit estimates

	Average derivative
	(1)
Panel A. Candidate observables	
Total publications	0.153***
	(0.003)
High-impact publications	0.095***
	(0.002)
Citations	0.073***
	(0.003)
Panel B. Exogenous shifters	
Coauthor	0.185***
	(0.035)
Same affiliation	0.162***
	(0.031)
Research Similarity	0.030***
	(0.006)
Observations	69,015
Chi-squared for test of excluded instruments	100.03

Notes: The sample includes all candidates who applied in the first wave of the 2012 ASN cycle. The table reports average marginal effects from a logit model of qualification on candidate characteristics and connections to the committee. Candidate research measures are standardized within examinations. The specification includes examination fixed effects. Standard errors are clustered at the examination level. The chi-squared test assesses the joint significance of the exogenous shifters.
*** p < .01, ** p < .05, * p < .1

Table D2: Effects of qualification on subsequent outcomes

	(1) Total Publications	(2) High-Impact Articles	(3) Citations (All Articles)	(4) Citations (Pre-Exam Articles)	(5) Citations (Post-Exam Articles)	(6) Promoted Associate Prof. (within 5 years)	(7) Promoted Associate Prof. (within 10 years)	(8) Promoted Full Prof. (within 5 years)	(9) Promoted Full Prof. (within 10 years)
Panel A. Associate Professor Candidates									
Qualified	0.657**	0.596	0.272	-0.056	0.516	0.479***	0.379**	0.035	0.091
	(0.327)	(0.394)	(0.362)	(0.231)	(0.440)	(0.151)	(0.163)	(0.025)	(0.111)
Observations	47424	47424	47424	47424	47424	47424	47424	47424	47424
R^2	0.12	0.13	0.33	0.61	0.10	0.14	0.04	-0.02	0.05
Panel B. Full Professor Candidates									
Qualified	0.681	0.718	0.781	-0.039	1.120*			0.292*	0.036
	(0.429)	(0.452)	(0.499)	(0.279)	(0.606)			(0.159)	(0.210)
Observations	21589	21589	21589	21589	21589			21589	21589
R^2	0.18	0.15	0.31	0.65	-0.04			0.10	0.07

Notes: The table reports 2SLS estimates of the effects of qualification on subsequent outcomes, instrumenting qualification with realized connections net of their expectation under the assignment mechanism. The sample includes applicants in the first wave of the 2012 ASN cycle: associate-professor applicants in Panel A and full-professor applicants in Panel B. Outcomes are measured ten years after evaluation unless otherwise stated. *Promoted Associate Prof.* and *Promoted Full Prof.* indicate promotion to the respective rank. Research outcomes are standardized within examinations. All specifications control for four broad disciplinary groups; candidates' pre-examination publication counts, publication impact, and citations; and affiliation and academic rank indicators. Standard errors are clustered at the exam level.
*** p < .01, ** p < .05, * p < .1